Swarm and Evolutionary Computation 53 (2020) 100631

Contents lists available at ScienceDirect

Swarm and Evolutionary Computation

journal homepage: www.elsevier.com/locate/swevo

Role of swarm and evolutionary algorithms for intrusion detection system: A survey

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ARTICLE INFO

Keywords:
Intrusion detection system
Challenges
Swarm and evolutionary algorithms
Future directions
Genetic algorithm
Ant colony optimization
Particle swarm optimization
Artificial bee colony
Firefly algorithm
Bat algorithm
Flower pollination algorithm

ABSTRACT

The growth of data and categories of attacks, demand the use of Intrusion Detection System(IDS) effectively using Machine Learning(ML) and Deep Learning(DL) techniques. Apart from the ML and DL techniques, Swarm and Evolutionary (SWEVO) Algorithms have also shown significant performance to improve the efficiency of the IDS models. This survey covers SWEVO-based IDS approaches such as Genetic Algorithm(GA), Ant Colony Optimization(ACO), Particle Swarm Optimization(PSO), Artificial Bee Colony Optimization(ABC), Firefly Algorithm(FA), Bat Algorithm(BA), and Flower Pollination Algorithm(FPA). The paper also discusses applications of the SWEVO in the field of IDS along with challenges and possible future directions.

1. Introduction

The domain of network security comprises of diversified fields of research such as Intrusion Detection System (IDS), cyber forensics and investigation, securing access control mechanisms, risk management, and auditing, and system log analysis [1]. The term network security can be coined as the procedure of securing information generated through the network to ensure confidentiality, integrity, and availability of the system [2]. Thus, network security is one of the major challenges while dealing with malicious activities in wired or wireless networks. The network activities are studied by using various network tools such as “Wireshark” [3] that can be configured to intercept the network packets. The intruder extracts the required network information from the captured network packets to execute network attacks. The network attacks being executed inside the network are categorized as internal network attacks while the ones performed outside the network are called external network attacks. These internal and external network attacks can be detected by IDS deployed in the underlying network. It protects the system from malicious activities by generating alerts.

Various methods implemented for detecting intrusions such as rule mining, neural network, classification techniques, clustering techniques, and other Deep Learning (DL) based approaches. These techniques were implemented to ensure the security of the network by detecting attacks with high accuracy. To evaluate the performance of an IDS, a suitable technique needs to be identified for the application specific dataset. The detection and classification of attacks performed by studying the features of the chosen dataset. For instance, a hybrid technique for feature selection is proposed in Ref. [4]. Here, chi-square feature selection method is used with enhanced C4.5 classifier to build efficient network-based IDS. To build an efficient IDS model, the proposed method selects maximum suitable features that can contribute in classifying attacks. A technique based on improved support vector decision function is implemented in Ref. [5] for selecting features. Here, feature selection is performed by assigning rank or weight to the features. This technique results in reduced training and testing time. A technique based on the artificial immune system is proposed in Ref. [6] for agent-based IDS. Though ML and DL techniques implemented for building an efficient IDS, these techniques exhibit challenges. An IDS designed for classifying attacks possesses characteristics which are not well-aligned with the requirements of ML and DL techniques. These characteristics include detecting the outliers, high cost of errors, the semantic gap between the results and their interpretation, diversity in the generated network traffic, and difficulty in evaluating the data [7].

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https://doi.org/10.1016/j.swevo.2019.100631
Received 24 May 2019; Received in revised form 7 December 2019; Accepted 9 December 2019
Available online 13 December 2019
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Abbreviations

ABC	Artificial Bee Colony
ACO	Ant Colony Optimization
ACS	Ant Colony System
ADR	Attack Detection Rate
AFS	Artificial Fish Swarm
AUC	Area Under Curve
BA	Bat Algorithm
BABC	Binary Artificial Bee Colony
BACO	Binary Ant Colony Optimization
BBA	Binary Bat Algorithm
BFA	Binary Firefly Algorithm
BPSO	Binary Particle Swarm Optimization
CC	Correlation Coefficient
CFPA	Chaotic Flower Pollination Algorithm
CPE	Cost Per Example
CR	Classification Rate
DBI	Davies Bouldin Index
DL	Deep Learning
DR	Detection Rate
DT	Decision Tree
FA	Firefly Algorithm
FAR	False Alarm Rate
FHC	Firefly based Harmonic Clustering
FN	FIN Flag
FNR	False Negative Rate
FPA	Flower Pollination Algorithm
FPR	False Positive Rate
GA	Genetic Algorithm
GNP	Genetic Network Programming
GP	Genetic Programming
HBA	Hybrid Bat Algorithm
HFA	Hybrid Firefly Algorithm
HG-GA	Hypergraph-Genetic Algorithm
HR	Hit Rate
ICS	Improved Cuckoo Search
IDS	Intrusion Detection System
KDD	Knowledge Discovery Database
KPCA	Kernel based Principal Component Analysis
MAE	Mean Absolute Error
MCC	Matthew's Correlation Coefficient
MLP	Multi Layer Perceptron
MND	Malicious Node Detection
NB	Naïve Bayes
NFR	Negative False Rate
NIDS	Network-based Intrusion Detection System
NN	Neural Network
NPV	Negative Prediction Value
PCA	Principal Component Analysis
PFR	Positive False Rate
PPV	Positive Prediction Value
PSO	Particle Swarm Optimization
RAE	Relative Absolute Error
RBF	Radial Basis Function
RF	Random Forest
RMSE	Root Mean Square Error
RN	Reset Flag
ROC	Receiver Operating Characteristics
RSE	Relative Standard Error
SD	Standard Deviation
SGD	Stochastic Gradient Descent
SIP	Session Initiation Protocol
SN	SYN Flag
SVM	Support Vector Machine
SWEVO	Swarm and Evolutionary
TPR	True Positive Rate

Therefore, to address the challenges Swarm and Evolutionary (SWEVO) algorithms can be used by hybridizing it with ML or DL technique for feature extraction or parameter optimization. Also, the integration of SWEVO algorithms with ML or DL technique can address huge network traffic, diverse data distribution, and continuously changing environmental conditions.

The benefits of SWEVO algorithms over ML and DL techniques are as follows.

- The dependence of SWEVO algorithms on the known labels within the dataset is far less compared to ML and DL techniques [8].
- These algorithms can be applied to the dataset for the classification right from the beginning without applying data pre-processing methods [9].
- These algorithms require less prior knowledge about the data and need minimum human expertise [9].
- Hybridization of SWEVO algorithms with ML and DL techniques result in improved performance of the model [7].
- SWEVO algorithms are inherently distributed and can work well with parallel and distributed computing systems [10].
- Models built using SWEVO algorithms are highly interpretable, flexible, and are capable of giving optimal solutions [8].
- SWEVO algorithms are capable of addressing multi-objective problems and are conceptually simple and easy to use [11].

A comparison between ML, DL, and SWEVO algorithms is presented in Table 1.

SWEVO algorithms applied to a variety of problem domains such as classification, regression, clustering, and optimization [13]. Moreover, these algorithms can be applied to a variety of applications, including developing game strategies, automation in program development, and optimizing the design for disparate domains [14]. These algorithms gradually emerging as a potential technique to study the complexity of computing the data for IDS [14]. This motivates us to have state-of-art literature on SWEVO algorithms applied for IDS. Therefore, this paper surveys SWEVO algorithms used for intrusion detection, parameter optimization, and feature selection. The prominence of these techniques geared the heights because of the way by which, a massive amount of the data handled by the algorithms [6]. The learning ability and adaptability of these algorithms resembles that of biological organisms. Therefore, these algorithms became the center of attraction for researchers working in different domains including network security.

1.1. Related work

SWEVO algorithms are implemented with the sufficient element subset and features for enhancing the performance of IDS [5]. Various types of IDS and its components are discussed in Ref. [15] along with the use of GAs in the field of IDS. The study discusses different work performed in the field of IDS for attack classification using GA. It also discusses how the parameter and membership functions can be tuned to enhance the process of intrusion detection. Techniques such as ACO and PSO along with artificial neural networks, fuzzy logic, and artificial immune systems are presented in Ref. [16]. The survey summarizes and compares each of the methods to derive the research challenges in the field of IDS and potential research directions. The different scenarios of Session Initiation Protocol (SIP) based flood attacks using different evolu-

Table 1
Comparison of MLDL and swarm and evolutionary algorithms.

Criteria	ML and DL	Swarm and Evolutionary Algorithms
Learning Paradigm	Statistical learning algorithms [7]	Global optimization algorithm [11]
Working Principle	Works at establishing a relation between output and input variables and subsequently developing a model [7]	Works at deriving the optimal and best feasible solution from given population [9]
Output	The output of ML and DL techniques would be a model subjected to input constraints [7]	The output of SWEVO algorithms would be the best output value subjected to input constraints [9]
Goal of Generalization	Aims at minimizing the computational cost of unseen data samples [7]	Aims at minimizing the computational cost of known data samples [12]
Applicability	These are used for classification, clustering, and regression problems [7]	These are used for classification, feature extraction, and parameter optimization [8]

Table 2
Comparison of surveys studied for IDS.

Ref	GA	DE	ACO	PSO	ABC	FA	BA	FPA	Applications in IDS
[15]	✓								
[16]			✓	✓					
[17]	✓								
[18]	✓								
[19]	✓	✓	✓	✓		✓			
[20]	✓			✓					
[22]	✓								
[23]	✓		✓	✓					
[8]	✓	✓	✓	✓	✓	✓			
[21]			✓	✓	✓	✓	✓		
[24]	✓								
Our Survey	✓	✓	✓	✓	✓	✓	✓	✓	✓

tionary and non-evolutionary algorithms are analyzed in Ref. [17]. The use of evolutionary algorithms for enhancing the performance of intrusion detection in wireless networks and resource-constrained networks presented in Ref. [18] including various research areas and challenges in IDS and GA-based IDS. In Ref. [19], an overview of SWEVO algorithms for feature selection is presented. The survey summarizes the algorithms with their advantages, disadvantages, and applied research areas. A survey on feature selection and classification methods for IDS such as neural networks, GAs, neuro-genetic techniques, fuzzy logic, and PSO is presented in Ref. [20].

A survey on swarm intelligence algorithms for feature selection and feature reduction is presented in Ref. [21]. The survey presents, guidelines to apply these algorithms for feature selection. The survey also discusses the existing research issues and open research questions. The components and functionalities of IDS along with the use of GA in IDS for mobile ad hoc networks is presented in Ref. [22]. A survey on GA, ACO, and PSO algorithms implemented for IDS is presented in Ref. [23]. Here, the paper presents a discussion on GA, PSO, and ACO algorithms along with their variants used for intrusion detection. A brief overview concerned with how bio-inspired and non-bio inspired algorithms applied for the feature selection process is presented in Ref. [8]. SWEVO algorithms can be hybridized with other techniques to improve the results of the problem under study. A case study to explain the same is presented in Ref. [24] where evolutionary algorithms are hybridized with fuzzy systems. A comparison of surveys studied for IDS is shown in Table 2.

For the identified SWEVO algorithms, our paper aims to.

- describe the mechanisms and conceptualization
- discuss the algorithms implemented in the field of IDS
- presents a review of different application areas of IDS
- list various challenges and research findings for IDS with potential future directions

The roadmap of the paper is presented in Fig. 1.

2. Introduction to Swarm and Evolutionary Algorithms

SWEVO algorithms are the promising research area of computational science that derives its inspiration from the natural evolution of the biological organisms [13]. Swarms can be described as the batch of individuals that are present in massive numbers and exhibit the same behavioural characteristics. The members of the swarm communicate with each other and the environment to converge to a global solution for the underlying application. Swarm intelligence algorithms built on the behavioural models of living organisms such as ants, birds, fish, and bees [25]. These algorithms built on the adaptability, flexibility, and coordination characteristics of the species. The members of a swarm exhibit a particular pattern to be followed by every member of a population without any centralized authority for controlling the communication between the individuals present in the swarm. The communication between the members can be direct or indirect. For example, in bee colonies, the bee members of a swarm communicate the information of the food sources through waggle dances. This form of communication is called direct. On the other hand, if the information is communicated to the members of a swarm by observing the change in the way other members reciprocate to the environmental changes is called indirect [25]. All the individuals of the swarm have a very low intelligence individually, but when they communicate with each other and with the environment, they can converge to an optimal solution. These algorithms can be used to address problems that are NP-hard such as traveling salesman, vehicular routing, scheduling optimization, and intrusion detection [26].

Evolutionary algorithms are biologically inspired population-based optimization algorithm [27]. These methods mimic biological processes such as reproduction, mutation, crossover, and selection. Agents in the swarm serve as a candidate solution for the given problem to be optimized and fitness of each agent measured using the evaluation function. The solution to the problem to be optimized is obtained by applying the variation operators. Evolutionary algorithms do not assume the fitness

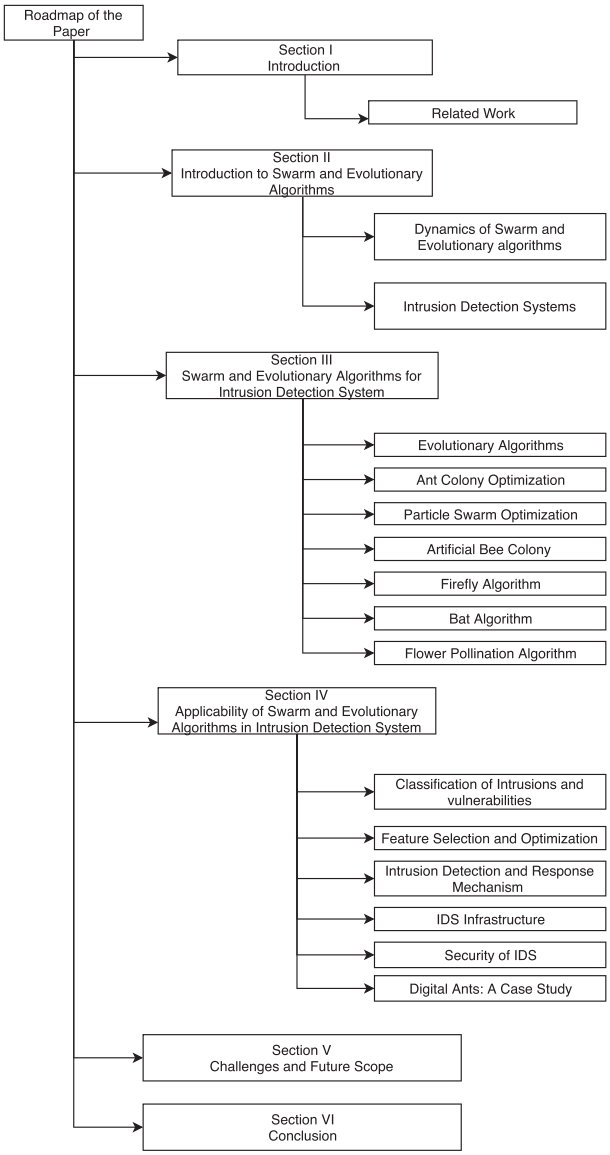


Fig. 1. Roadmap of the paper.

parameters and therefore, good approximating solutions are obtained for the given problem [9]. A comparison of evolutionary algorithms and swarm intelligence algorithms is presented in Table 3.

Both swarm intelligence and evolutionary computation algorithms have been used in various applications [13]. For instance, SWEVO algorithms were defined for the application of the robotics system in Ref. [31]. The cellular robotics system contained free agents that interact with each other for a specific purpose. Also, a large number of nanobots mimicked the phenomenon of natural science by acting together in their physical environment as narrated in Ref. [32]. Moreover, many applications were addressed using these algorithms such as Ant Colony Optimization (ACO); was introduced in Ref. [33] to solve hard combinatorial optimization problems, Particle Swarm Optimization (PSO) was the first algorithm to reproduce the behaviour of flocking of birds in real-world applications [34]. Later, Artificial Bee Colony (ABC) algorithm was developed by taking inspiration from the intelligent interaction of honey bees [35]. Due to the computational efficiency of these swarm algorithms, optimization problems were solved successfully. The examples inspired from swarm algorithms are transport mechanism of ants has motivated the working and design of the controllers used for robot [36]; the progeny behaviour of ants were used for clustering and sorting algorithms [37–39]; the working of a collective set of robots was inspired from the work distribution among ants [40]. The research performed in the field of IDS develop models for detecting intrusions by mimicking the interactions between the flocking of birds, school of fish, colonies of ants, foraging behaviour of bacteria, the attractiveness of the firefly, and obligatory behaviour of a cuckoo [9]. A taxonomy of SWEVO algorithms discussed in this study is presented in Fig. 2.

2.1. Dynamics of swarm and evolutionary algorithms

SWEVO algorithms can be regarded as the exposure of intelligent, collective, self-configurable and possessing global characteristics algorithms [41]. This section discusses the general principles and components of SWEVO algorithms in brief.

2.1.1. Principles of swarm and evolutionary algorithms

To model the behaviour of the individuals present in the SWEVO algorithms, general principles can be deduced which are listed as follows [9,11].

- Proximity Principle: The agents of the swarm should possess the ability to respond to small environmental changes through their behavioural response and interact with the other members of the swarm. The response made against the variance caused by the environmental changes highly depends upon the total number of agents involved. The behavioural response can be of any form such as searching for potential resources [9,11].
- Quality Principle: Apart from responding to the environmental variance, the members of the swarm should be capable of judging the quality factors like location safety, availability of the food resources, and finding trails for the correct paths towards the resource [9,11].
- Principle of diverse response: The resources should be distributed in such a manner that every member of the swarm has enough resources to protect itself at the time of environmental changes. They should not be focussed on the narrow region of the population [9,11].
- Principle of Stability: The members of the swarm should possess the capability of maintaining the behavioural patterns at the time of the environmental fluctuations [9,11].
- Principle of Adaptability: The members of the swarm should possess the capability of easily adapting to the new swarm behaviour when the mode of the environment changes. As the change in the mode of environments requires a high energy cost [9,11].

2.1.2. Mechanisms of swarm-based algorithms

The swarm-based algorithms comprised of environment and agents. These agents interact with each other in the environment and converge to a common solution for a problem [26]. Each of these mechanisms is described as follows.

- Environmental Mechanism: Here, the members of the population communicate with each other via some mode of actions such as waggle dances, pheromone trails, or intensity of the light to transfer the necessary information to the other members of the swarm. This serves as an external memory where information is passed through simulation [25].
- Interaction Mechanism: The interaction between the agents can be direct or indirect. For instance, simulating the information of environmental changes by performing actions is indirect interaction and intimidating the other members about the available resources is direct interaction [25].
- Activities of agents: The action performed by the agents is based on the random process that maintains a balance between the assumptions and reactions made by the agents. The individual receives information about the local properties of the environment through these assumptions and reactions. Apart from the interaction, the

Table 3
Difference between evolutionary algorithms and swarm intelligence algorithms.

Criteria	Evolutionary Algorithm	Swarm Intelligence
Conceptualization	Generic population based algorithms inspired by biological evolution.	Self-organized, decentralized algorithms based on the collective behaviour agents.
Solution Ranking	They rank the solutions based on the fitness value and eliminates the solution with least fitness value [28].	They select the best solutions among the available solutions [28].
Impact of Population Size	The solution time increases exponentially based on the population size [29].	The solution time increases linearly based on the population time [29].
Premature Convergence	The tendency of premature convergence is moderate [29].	The tendency of premature convergence is high [29].
Search Space	The density of search space is low [30].	The density of search space is high [30].
Examples	GA, GNP, GP, Differential Evolution, Evolutionary Strategy.	ACO, PSO, ABC, FA, BA, FPA are a few examples of swarm intelligent algorithms.

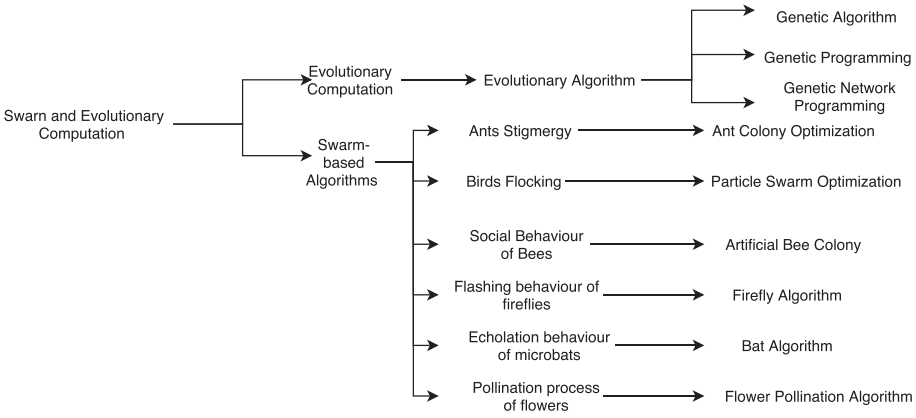


Fig. 2. Taxonomy of swarm and evolutionary computation.

agents can randomly move in the surrounding when no information needs to be exchanged between the agents [25].

2.1.3. Mechanism for evolutionary algorithms

The evolutionary algorithms comprised of an environment where the individuals are formed by mechanisms inspired by biological evolution [27]. The individuals try to form a solution that is better than the solution available. Each mechanism is described as follows.

- Generation of Individuals: This is the process by which candidate solutions created from the already available parent solutions. This is necessary when parent solutions cannot be improved further [27].
- Mechanism of Variation and Reproduction: This mechanism describes the formation of possible solutions with an appropriate difference between the parent solution and the child solution. The reproduction and variation process is carried out using various variation operators such as crossover and mutation [27].
- Mechanism of Selection: The individual's fitness is calculated using the fitness function and the individual with the high fitness value is selected for producing the candidate solutions and the ones with the low fitness value are abandoned [27].

2.2. Intrusion detection systems

Many techniques implemented for securing computer systems such as cryptography, firewall, network security protocols, and access control mechanisms. With the evolution in network traffic, the types of attacks over the course of time have changed substantially, and therefore, there is a high need for a revolution in the security methods to be adopted for securing the computer systems. The network architecture can be of any form such as cellular network [42], wireless sensor networks [43], vehicular ad hoc networks [44], mobile ad hoc networks [45], Bluetooth technologies [46] and 4G technologies [47] which are capable of generating large amount of data and hence, increase the potential vulnerabilities. The conventional security mechanisms such as firewall and access control mechanisms lack the potential to address the intrusions caused in the defined network and thus, IDS can be stated as the first line of defense in the security infrastructure [48]. An IDS is a mechanism to examine the network traffic and analyze the behavioural patterns for protecting the computer systems. The most promising and challenging part is to build an IDS which is efficient enough to detect any type of attack probable to occur in the network. An IDS exhibiting a high Detection Rate (DR) and low False Alarm Rate (FAR) can be considered as the ideal one [14].

An IDS model was presented in Ref. [48], and since then, various techniques have been introduced for increasing the efficiency and effectiveness of IDS. Classification of IDS is presented in Fig. 3 [49]. An IDS can be classified based on attributes such as information source, analysis strategy, locality, response mechanism, and decision making [49]. The most important task of any IDS is to cope up with constantly changing the real-time behaviour of the networks. For evaluating the performance of IDS, the intrusion detection problem is considered as the classification problem that can be solved by applying classification and/or clustering algorithms to the dataset of IDS. The identified model learns different patterns and signatures from the records during the training phase. While during the testing phase, the test records are classified by

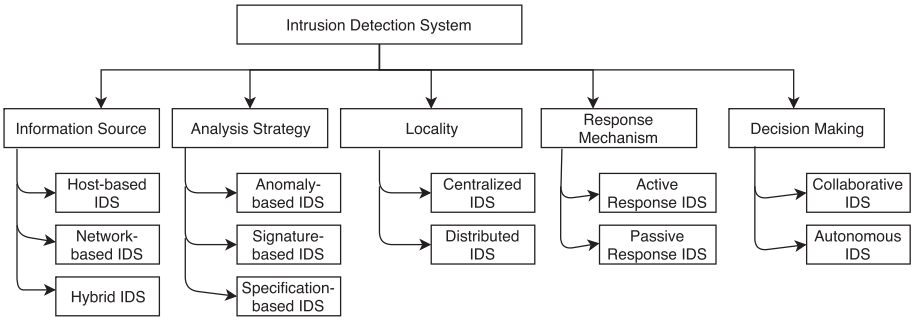


Fig. 3. Classification of intrusion detection system [49].

the trained model. The classification methods are easy to implement, quick, and efficient [49].

To meet the requirement of efficiency and effectiveness in IDS, bio-inspired techniques have also been taken into consideration [50]. SWEVO algorithms can be used in IDS for distinguishing benign and malicious traffic, identifying the source of the attack, feature selection, and for optimizing the performance of the classifier. This is because these algorithms possess the property that can be easily inherited in the IDS. For example, consider the swarm of bees that can accomplish the tasks of detecting intrusions or optimizing the performance of classifier [51]. It can easily adapt to the environmental changes and cope up easily even if some members of the swarm exhaust. Similarly, any IDS built using the swarm-based methods depicts the characteristics of the swarm such as highly adaptive to changes, robust, self dependable, and simple to implement [9]. The unique characteristics of the swarm-based algorithms can be used for incorporating the classification rules in the IDS, identify clusters for detecting malicious activities, and trails of activities of the attacker can be tracked. By this, the intrusion detection problem can be divided into multiple simple problems that can be assigned to different agents. This potentially makes an IDS self-configurable and can reduce the cost of operation.

3. Swarm and Evolutionary Algorithms for Intrusion Detection System

SWEVO algorithms aim to gain better accuracy for a given problem and converge to a common goal by obtaining a set of solutions. These algorithms designed for a single objective and multi-objective optimization to obtain an optimum set of solutions [52,53]. These algorithms achieve the optimization stage by exploring the problem domain and computing the fitness function. Therefore, many algorithms developed by studying the behaviour of the group of individuals and applied for solving complex problems. These algorithms find an optimal set of solutions with low computational complexity [9].

For a given problem, SWEVO algorithms consist of components, functions, and operators for finding an optimal set of solutions. The goal of these algorithms is to form a connection between the real world and the evolutionary world. The components, functions, and operators for the given SWEVO algorithm are as follows [52].

- The objects identified as the probable set of solutions for a given problem are called as a representation or a group of individuals.
- For a given group of individuals, population refers to the number of agents present in the group.
- For addressing the problem in an evolutionary context, some actions are performed by the group of individuals which is called as the fitness function.
- The parents are selected for a given generation of the population based on the value obtained by the fitness function.
- The high-quality individuals are selected as the parent among the population and the rest individuals exhaust and die.
- The population for a given context problem should remain constant and therefore, new individual is created from the old ones by using the operators such as crossover and mutation.
- The process of obtaining an optimum fitness level continues until a suitable stopping condition is met.

3.1. Evolutionary algorithms

Evolutionary algorithms are population-based algorithms inspired by the biological process. The commonly used evolutionary algorithms are Genetic Algorithm (GA) and Genetic Programming (GP) [54]. GA was developed to imitate the natural procedure of creating off-springs using computation methods to achieve optimal solutions for the complex problems [55]. It is a heuristic evolutionary algorithm that imitates the procedure of natural selection using nature-inspired operators like crossover and mutation to obtain competent solutions for the defined problem context [55]. Evolutionary algorithms are used for solving problems that are single objective, multi-objective, combinatorial, and/or non-deterministic [54]. To apply GA, a fitness function is described and used to obtain a solution referred to as the fitness value that indicates the quality of the solution. The fitness value can be calculated from a set of integers, matrix, or any other method based on the description of the problem addressed by the algorithm. The fitness function can be either maximizing or minimizing fitness function based on the goal defined for the problem [56].

The operators used for GA are inheritance, crossover, reproduction, and mutation [57]. These operators are used for generating a set of new potential solutions from already available solutions that are pre-optimized. For example, crossover and reproduction operators combine the characteristics of the parent's solutions and produce a child solution with the most useful characteristics of the parent solution. While the mutation operator is used to increase the fitness value of the child solution produced from the crossover and reproduction operators. This ensures randomness and avoids local optimality of available solutions associated with the problem [58]. The GA is less efficient with the high dimensional, multi-modal and complex problems where evaluating the fitness values for the solutions is computationally very expensive and time-consuming [57]. The attributes of evolutionary algorithms are shown in Fig. 4.

The evolutionary algorithms are used for optimizing features or parameters of the classifier to classify attacks in IDS. The schematic model of evolutionary algorithm optimization is depicted in Fig. 5. The evolutionary algorithms consider the features of the given dataset as chromosomes and learn the importance of each feature by evaluating its fitness value. The selected features are then fed to the classification module, where different attack patterns are derived to perform classification. Evolutionary algorithms such as GA and DE have been used for IDS [19].

In Ref. [59], GA is applied for optimizing the fuzzy association rules that are used for mining the best features and deriving the correlation between extracted features. The proposed method implemented in

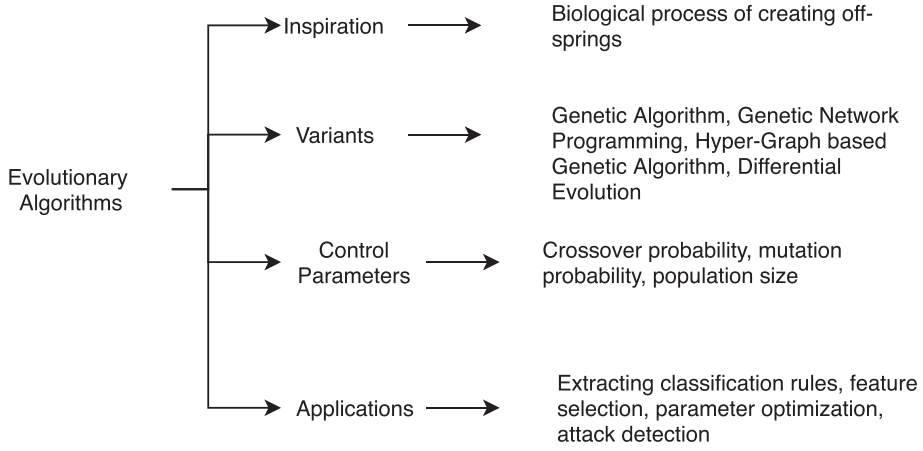


Fig. 4. Evolutionary algorithm.

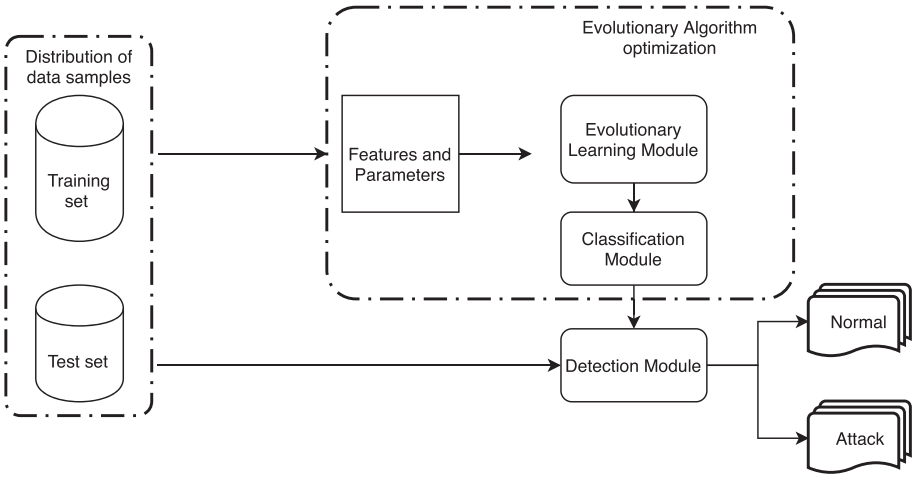


Fig. 5. Schematic model of evolutionary algorithm optimization.

the network containing 60 nodes and network packets captured using tcpdump. The features extracted from the TCP header packets consist of various flag information such as SYN, FIN, and RST. To study the network traffic for detecting intrusions, fuzzy association algorithm applied to these features and further these rules are optimized using GA. GA also used for resource-constrained wireless sensor network where detecting intrusions is quite challenging due to the limited capability of available resources. Genetic Network Programming (GNP) is used with fuzzy association rule mining to deal with a dataset containing continuous as well as discrete attributes [60]. GNP uses a directed graph structure to extract the best rules for detecting intrusive patterns in the data [60].

Generally, rule-based systems combined with GAs are referred to as learning classifier systems [61]. These systems have capability to adapt the environmental changes easily. Moreover, rule-based systems can extract and even update signatures in the database for intrusion detection [61]. Rule-based systems are even robust in detecting the zero-day attacks as they consistently update the signatures by evaluating the features using the fitness function of GA. In Ref. [61], each rule set is considered as a separate classifier. Each rule set consists of a condition and the corresponding action to be performed, and a set of parameters. The proposed system defines two major parameters namely accuracy and fitness. The accuracy is represented as the ratio of correctly classified rules to the total number of instances and fitness is evaluated as the direct function of accuracy. The experiments are performed on KDD CUP 99 dataset with 510 signatures of normal as well as malicious network traffic. Apart from updating the ruleset and the feature vectors, fitness function of GA can also be used for the actual detection process of the intrusion by assigning weights to the classes of the underlying dataset and matching the network patterns for intrusion detection [62]. In this way, both spatial and temporal data can be incorporated to identify intrusions for given network connections.

In Ref. [63], a network of 100 nodes is developed wherein a local monitoring node is configured to secure and monitor the neighbouring nodes against the malicious activities. This local monitoring node behaves as a trusted proxy agent. Here, GA optimizes the location of the local monitoring node by measuring the fitness value by analyzing workload patterns, packet header, packet payload statistics, usage of battery, and quality-of-service [63]. It should be noted that the IDS analyses the network traffic flows to extract the patterns at different time intervals and generates alerts if any unusual pattern is found. The GA can be used to generate signature from the flow analysis of the network system by extracting information from the network flow data. Based on the learned patterns, fuzzy logic classifier is used for representing a particular instance as anomalous or normal [64]. Here, the proposed system is used for traffic characterization of the captured packets and anomaly detection. The implementation uses numeric encoding for studying the behavioural patterns of the network data and identifying attributes from the data. It speeds up the analysis process and ensures real-time traffic analysis. The quality of attributes extracted from the data is evaluated using the fitness function. The Euclidean distance is used as the fitness function as shown in equation (1). Here, k

is the attribute indexer, y_k is the attribute value and x_k is the observed behaviour of the data sample. Thus, the attributes are selected based on their fitness value. The signatures and attributes generated using GA are fed as input to the fuzzy inference system for anomaly detection. The experiments are performed using KDD CUP 99 dataset and results expressed in terms of normalized mean squared error and ROC curve.

$$f_k = \sqrt{\sum_{i=1}^{n} (y_k - x_k)^2} \tag{1}$$

GA is also be used for optimizing the parameters of the classifiers to increase the efficiency of the classifier. GA is used for parameter optimization of Support Vector Machine (SVM) namely, the punishment factor C, kernel parameters σ and the tube size ϵ [65]. The proposed method improves accuracy and convergence rate. GA is also used with SVM for deriving features from the dataset that are important for classification. GA is used for feature selection in Ref. [66] wherein the selected features are prioritized by grouping them based on their importance. Thus, the features with the highest importance are grouped as a first priority, whereas features with the lowest importance are grouped as a third priority. These selected features are used by the SVM to classify and detect attacks. The paper [67] states that the existing IDSs poses the problem of fidelity, resource usage, and reliability. To address these issues, the paper uses KDD CUP 99 dataset for evaluating the IDS in which the total number of attacks in the dataset is represented as chromosomes and 23 chromosome groups were recorded to classify attacks and normal network traffic. In the detection phase, the set of chromosomes whose fitness value is closest to that of the fittest surviving chromosome of test data is regarded as the predicted type of attack. GA is used for selecting classification rules for predicting attacks in Ref. [68] wherein GA is used with Naïve Bayes (NB) and Decision Tree (DT) to select the best classification rules for the prediction. The quality of the classification rule is measured using fitness function as defined in equation (2). Here, a denotes correctly identified attacks, A denotes total number of attacks in the dataset, b denotes the number of false positives, and B is the number of normal instances in the dataset. GA is even used to optimize the false positive rate for intrusion detection model in Ref. [69] by selecting the classification rule for anomaly detection.

$$Fitness = \frac{a}{A} - \frac{b}{B} \tag{2}$$

The critical part of building an IDS is dependant on the processing of the network data, extracting significant features, and building an efficient model that can classify the malicious and non-malicious data [70,71]. For instance, in Ref. [72], the designing of IDS is performed in four phases: in the first phase network data is preprocessed and redundant data is removed. In the second phase, GA is used for feature extraction in which a 41-bit binary string is used to represent each individual chromosome in the population. The fitness of each chromosome is evaluated based on the specificity and sensitivity of the dataset. Feature subset is selected by applying uniform crossover, simple mutation, and tournament selection [72]. In the third phase, the obtained feature vectors are normalized as per the data smoothing function. In the last phase, neuro tree is used as a classifier where neural network is ensembled with a C4.5 classifier. The proposed algorithm was combined with other DT classifiers and showed an increase in the DR. The variant of GA named Hypergraph-GA is used for feature extraction in Ref. [73] where features are selected by traversing a graph and classification is performed by SVM. The selection of features is performed using a fitness function. The fitness function demonstrates the importance of detection rate, false alarm rate, and the number of features. The number of features that result in a high detection rate and low false alarm rate is selected. The fitness function is given in equation (3). Here, D_W, FA_W, and F_W are the weights defined for detection rate, false alarm rate, and features, respectively, and N is the total number of the features present in the dataset.

$$Fitness = D_W[DR] + FA_W[1 - FAR] + F_W\left[1 - \frac{\sum_{i=1}^{N} F_i}{N}\right] \tag{3}$$

Differential Evolution (DE) is used to detect anomalies as well as for selecting features in IDS. It is based on the concept of utilizing a large population and applying self mutation for obtaining candidate solutions [74]. DE generates new candidate solutions by incorporating a weighted difference vector between individuals. In Ref. [75], feature reduction is performed using a flexible neuro tree that extracts important features from the dataset. Thereafter, rules for classifying attacks are evolved using DE. The extracted features are represented using a continuous function. The quality of each element is measured using a fitness function given by equation (4) wherein $|A + B|$ is the total of network connection that is correctly identified, sim_z are the data samples that are partially matched by the rule, N_{normal} is the total number of normal instances and D is a constant and its value is 5 [75]. The proposed approach is applied to KDD CUP 99 dataset. The experimental results showed a good DR and FAR.

$$f(x_z) = \frac{|A + B|}{N_{normal}} + \frac{sim_z}{N_{normal}^2 * D} \tag{4}$$

In the study [76], a comparative analysis between a set of binary SVM classifiers is presented. The paper aims to build a multi-class SVM model that is capable of detecting intrusions from the given IDS dataset. Apart from comparing the binary SVM classifiers, a new method based on a weighted one against rest SVM is proposed that uses DE with iterative search to find the optimal set of SVM parameters. The parameters are selected by considering overall accuracy as the fitness function. The proposed method reduces the error in predicting the intrusions and also builds multi-class hypotheses where every binary classifier represents a specific set of classification parameters. The experiments are performed on NSL-KDD dataset. The results showed that the proposed method outperforms the variants of binary SVM classifiers in terms of classification accuracy.

DE is also used for selecting features from the dataset for improving the performance of the IDS model. In Refs. [77,78], DE is used as feature selection method. In Ref. [77], a feature selection method is proposed using DE and is applied to both training and test set. The classification is performed using DT classifier and the results are compared with other ML techniques. The proposed method uses overall accuracy as the fitness function to signify the quality of the selected features. The proposed method extracted 16 features from the 41 features defined in NSL-KDD dataset and achieved 99.92% Classification Rate (CR) on the training set and 88.73% CR on the test set. The study presented in Ref. [78] shows a comparative analysis of GA, PSO, and DE on DARPA dataset. All three methods are used for selecting features while classification is performed using SVM and NN. GA, PSO, and DE extract 16, 15, and 31 features, respectively from a total of 41 features available in the dataset. The overall comparison of CR shows that DE outperforms GA and PSO. Table 4 summarizes the role of GA, GNP, GP, and DE in the field of IDS.

Following is the list of advantages of evolutionary algorithms.

- Evolutionary algorithms exhibit intrinsic parallelism characteristic [59].
- These algorithms are capable of handling multi-dimensional data and hence, large volumes of intrusion detection dataset can be processed easily [60].
- These algorithms possess a population of solutions rather than a single solution. This characteristic of evolutionary algorithms can be easily implemented for behaviour-based IDS where behaviour profiles of users in a network are considered for detecting intrusions [66].

Table 4
Evolutionary algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[59]	Rule Extraction	Not specified	Fuzzy Rule Mining - GA	Simulated Dataset	60 nodes	19% of the training cases recorded a confidence value of 86.3% that showed value very low in case of SN and RN while low in case of FN	• Intrusion detection at the time of heavy network traffic is low. • The system is capable of identifying the patterns for users at different time slots.
[63]	Local monitoring nodes selection	Uniform crossover, roulette-wheel selection	GA	Stimulated Dataset	100 nodes	• MND varies from 0.05 to 0.30 • FPR ranges from 2.5% to 5%	• The proposed method results in faster detection of compromised nodes compared to traditional intrusion detection mechanisms. • The increase in the number of nodes results in early convergence of GA.
[61]	Signature extraction and updation	Stochastic selection, uniform crossover, simple mutation	Rule Based - GA	KDD CUP 99	Training set: 5 million records, Test set: 0.3 million	• Overall Test Accuracy: 92% • FAR: 0.84% • HR: 90.72% • CPE: 0.26	• The proposed framework is for an automatic and adaptive extraction of signatures. • The method requires a large amount of CPU time for processing.
[62]	Rule extraction	GA evaluation function based on weights	Fuzzy Logic - GA	DARPA	Not specified	Data representation and parameters of GA and their effectiveness on intrusion detection are discussed.	• The proposed method considers both temporal and spatial network information for encoding • Aggregating the information from the network sensors is a challenging task.
[60]	Rule updation and attack classification	GNP association rules are evolved using random selection, crossover and mutation	Rule Mining - GNP	KDD CUP 99	Training set: 3342, Test set: 990	For Misuse Detection: • DR: 98.7% • PFR: 0.53% • NFR: 3.75% For Anomaly Detection: • DR: 95.9% • PFR: 10.3% • NFR: 2.1%	• The problem of overfitting of data is removed by defining sufficient rules for each class.
[72]	Wrapper-based feature selection	Crossover probability: 0.7, Mutation probability: 0.001 and tournament selection	NeuroTree Classifier - GA	KDD CUP 99	Training set: 5 million records, Test set: 0.3 million	• Kappa Statistic: 0.9729 • MAE: 0.0018 • RMSE: 0.0364 • RAE: 3.34% • RRSE: 22.04% • DR: 98.38% • FAR: 1.61%	• The performance of the model depends on training data, technique used, and selected features. • The proposed method is suitable for multi-class classification and detection of specific attack.
[67]	Attack detection and classification	Crossover rate: 0.15, Mutation rate: 0.35 and tournament selection	GA	KDD CUP 99	Training set: 494,021 data samples, Test set: 311,029 data samples	• DR: 0.95 • FPR: 0.304	• The proposed method resulted in high false positive rate and considerable detection rate. • The proposed method has ignored the non-numerical features and therefore, it does not precisely detects the normal network traffic.

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Table 4 *(continued)*

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[68]	Rule selection	single point crossover, mutation rate: 0.01 and tournament selection	DT-NB-GA	KDD CUP 99	Training set: 100,000 data samples Test set: 976 data samples	• DR: 95.72% • FAR: 4.27%	• The system uses only three network features for attack detection and classification. • Representation of rules with GA, results in faster detection of attacks.
[78]	Feature selection	Greedy selection, Crossover probability: 0.7, Mutation probability: 0.024	GA-SVM and NN	KDD CUP 99	Training set: 494,021 data samples, Test set: 311,029 data samples	Accuracy convergence curve for SVM and NN using GA is presented.	• The proposed method results in high execution time. • Feature selection has resulted in improved classification accuracy and false positive rate.
[65]	Parameter optimization	Roulette wheel selection, single-point crossover probability: 0.8, mutation probability: 0.05	SVM-KPCA-GA	KDD CUP 99	Training set: 49,920 data samples, Test set: 60,254 data samples	• DR: 96.03% • FAR: 0.98% • CC: 0.963	• Feature extracted using kernel methods improves the generalization performance of the algorithm. • The hybridization of feature selection using KPCA and parameter optimization using GA results in less processing time.
[75]	Attack classification and detection	Simple selection, binomial crossover, single vector difference mutation operator	DE	KDD CUP 99	Training set: 494,021 data samples, Test set: 311,029 data samples	• DR: 100% • FAR: 0.528%	• The proposed method was able to reduce the features and classify the benign and attack traffic. • The performance of proposed framework is not evaluated for multi-class.
[69]	Optimization of false positive rate	Roulette wheel selection, two point crossover, swap mutation	GA	KDD CUP 99	Training set: 494,021 data samples, Test set: 311,029 data samples	• DR: 96.39% • FAR: 1.20%	• The proposed work show that false alarm rate is inversely proportional to the number of rules. Increase in the number of rules for detection, decreases the false alarm rate.
[70]	Feature selection	Randomized selection, uniform crossover, simple mutation	NB-GA	NSL-KDD	Traning set: 125,973 data samples Test set: 22,544 data samples	• DR: 96.96% • FAR: 0.22%	• The feature selection process has resulted in reduction in time complexity.
[71]	Feature extraction	Crossover rate: 0.6, mutation rate: 0.033	Sequential GA and Parallel GA	Spambase, NSL-KDD, Kyoto, CDMC2012	Spambase: 4601, NSL-KDD: 5000, Kyoto: 5000, CDMC2012: 10,000 data samples	Run-time analysis, and MapReduce implementation improvement analysis based on the number of attributes and number of instances is shown.	• The experiments show that parallel implementations have more consistent results compared to sequential GA. • The performance of the proposed framework is affected by the number of attributes and the number of instances used for implementation.

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Table 4 *(continued)*

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[66]	Feature selection	Two point crossover rate: 0.6, bit flip mutation rate: 0.001	GA-SVM	NSL-KDD	Not mentioned	• TPR: 0.973 • FPR: 0.017	• The evolution of features using GA technique has improved the false positive rate and true positive rate.
[73]	Parameter optimization and feature selection	Tournament selection, crossover rate: 0.8, mutation rate: 0.02	SVM-GA	NSL-KDD	Traning set: 16,613 data samples Test set: 196,110 data samples	• DR: 97.14% • FAR: 0.83%	• Hyper – clique property of hyper-graph decreases the complexity of GA. • The proposed method is scalable and adaptive over wide range of data analytical problems.
[76]	Parameter Optimization	Crossover probability: 0.5, differential weight: 0.8	DE-SVM	NSL-KDD	Training set: 125,973 data samples, Test set: 17,019 data samples	• Accuracy: 80.65%	• Model selection using DE, improved the multi-class classification. • An appropriate feature selection techniques is needed for improving the accuracy of the model.
[77]	Feature selection	Crossover rate: 0.5, random mutation vector	DE-DT	NSL-KDD	Training set: 125,973 data samples, Test set: 22,544 data samples	• CR on training set: 99.8% • CR on test set: 88.7%	• The implementation of feature selection resulted in decrease in training time.
[64]	Digital signature generation and network analysis	Binary tournament selection, crossover rate is mean between selected chromosomes, mutation is 3% chance to add or subtract 1% of the chromosome value	Fuzzy Logic-GA	Real Traffic	Not mentioned	• Accuracy: 96.53% • FPR: 0.56%	• The proposed method is an autonomous, and does not require a labeled data for detecting anomalies.

- Evolutionary algorithms can be re-trained. Therefore, adaptability characteristics can be achieved by applying these algorithms for detecting attacks in variable environmental conditions [74].
- The controlling parameters of evolutionary algorithms such as crossover probability and mutation probability evolve over time and therefore, these algorithms are suitable for extracting rules for intrusion detection [59].

Though evolutionary algorithms are efficient for building IDS models, they also have certain limitations.

- Formulation of fitness function for evolutionary algorithms is a challenging task [9].
- The representation of search space and feature space for IDS is complex using evolutionary algorithms [78].
- There is a probability of premature convergence to a solution in evolutionary algorithms [9].
- Choosing appropriate values for control parameters is also a challenging task [79].
- To effectively choose solutions from search space, an appropriate local search technique is required [78].
- These algorithms require a large number of fitness function iterations for converging to a solution [9].

3.2. Ant Colony Optimization

ACO is a heuristic search algorithm based on the communication of ants that interact with each other to find an optimal food source. The interaction among the ants is often indirect and the information regarding food source is communicated by pheromone trails [80,81]. These trails serve as a medium to develop solutions based on the probabilistic search carried out by the agents based on the distributed numerical data provided to them. The computation time to derive the optimal set of solutions is very less and is obtained in an iterative manner [82–84]. The candidate solutions prepared from already available solutions by performing parametric probability distribution. To derive the candidate solution, a group of agents is selected to choose the design variables for developing the solutions. The agents carry out the exploration and exploitation of the solutions. The solutions are updated based on the quality of the previous solutions. The updated solutions are used to modify the trails so that a good quality solution is selected by the group of agents. Random solutions considered during the initialization of the algorithm paves the path for obtaining the optimal set of solutions [80]. The attributes of ACO is shown in Fig. 6.

The challenging task in IDS is to protect the system from anomalous behaviour and intrusions. Various approaches have been implemented to build IDS models but model maintenance is a tedious task since the intrusion patterns and network attacks need to be examined precisely and respective rules need to be derived for generating the alert. To ease the task of attack classification a multiple level hybrid classifiers is proposed in Ref. [85] where DT classifier is combined with ACO. Here, ACO is used as the clustering algorithm for grouping the mixed data samples into clusters. The major advantage of applying ACO for clustering the data was to handle the unlabelled data efficiently [85]. The method proposed in Ref. [85] also has the capability of classifying unknown attacks. The results were obtained for KDD CUP 99 dataset that showed a reasonably high DR and low FAR. ACO is also applied for improving the classification rules for classification of attacks. A hybrid evolutionary fuzzy system proposed in Ref. [86] that involves two stages. In the first stage, fuzzy classification rules are defined using a fuzzy rule set for pattern classification. In the second stage, ACO is used to improve the performance of one fuzzy classifier at a time using an incremental learning pattern to enhance the accuracy of the proposed system. The fitness function used for generating fuzzy rules is given in equation (5). Here, w_p is the weight of positive classification and w_N is the weight of negative classification, f_p is the rate of positive training instances identified by the rule and f_N is the rate of negative training instances identified by the rule.

$$fitness = w_p f_p - w_N f_N \tag{5}$$

The proposed approach addressed the two problems in intrusion detection that are defining the classification rule and tuning the rule for achieving better classification accuracy. The experiments have been performed using the DARPA dataset and results have been compared with the other state-of-art methods.

A firewall security system also defines a ruleset for filtering the network traffic flowing in and out of the network [87]. Here, the packet filtering technique based on NN does not accept the packets that are not the part of packet acceptance ruleset defined for the firewall. Moreover, the proposed method does not accept packets if they are sourced from the same IP address or port multiple times [87]. This drawback is addressed in Ref. [88] where ACO based packet filtering approach is proposed. The proposed method considers the packets coming from the same source multiple times and is accepted or rejected by considering parameters defined in the firewall ruleset. The search space of the proposed method is smaller compared to the search space defined for the binary search [88]. The results of the defined technique are collated with the binary search technique, sequential search technique, and neural networks. A three-level ACO is proposed in Ref. [89], where ACO is used as a feature selection method. In the proposed work ACO searches for the optimal set of features in the feature space in an iterative manner based on the pheromone value of each iteration. The best feature set obtained is then validated using SVM to classify malicious and non-malicious network traffic. The experiments performed on KDD CUP 99 dataset and results compared with other feature selection techniques.

ACO is combined with the clustering algorithms such as k-means [90], harmonic k-means [91], and c-means [92] to form and initialize clusters. In Ref. [93], ACO based clustering method proposed with the kernel method. The main goal of the proposed technique is to improve the cluster quality and efficiency of the algorithm. The kernel method incorporated by considering two aspects: modification in the initialization of the data samples using the kernel-based PCA and measuring the similarity index of the data samples in the feature space using Euclidean distance. The experiments conducted on KDD CUP 99 [94] dataset and the training dataset is constructed using ACO along with k-means clustering. The feature set is reduced by using the genetic fuzzy rule mining approach. The work proposed in Ref. [95] aims at enhancing the efficiency and accuracy of the IDS model by pre-processing the data using the ACO algorithm and developing the feature selection approach for selecting the features from the feature set. The compact dataset is derived in two stages. In the first stage, k-means clustering used for refining the raw dataset by removing redundant records. In the second stage, the refined dataset is then fed as input to ACO. ACO algorithm along with SVM is applied to KDD CUP 99 dataset with 10-fold cross-validation to derive the subset of data with highest accuracy. Moreover, after the compact dataset is derived, a gradual feature reduction method is applied to the feature set for selecting relevant features. The experiments performed on KDD CUP 99 dataset. Though the proposed approach gave high detection accuracy, it did not work well with multi-class classification. Moreover, applying the proposed work on high dimensional datasets is a challenging task, especially in the case of an unbalanced dataset.

SVM and ACO based hybrid approach for self-organized clustering has been described in Ref. [96]. The proposed work aims at combining the advantage of classification efficiency of SVM and clustering efficiency of self-organized ACO. In Refs. [97–99], ACO algorithm is used with SVM as a feature selection algorithm to filter out insignificant features from the feature set. The performance of ACO for selecting

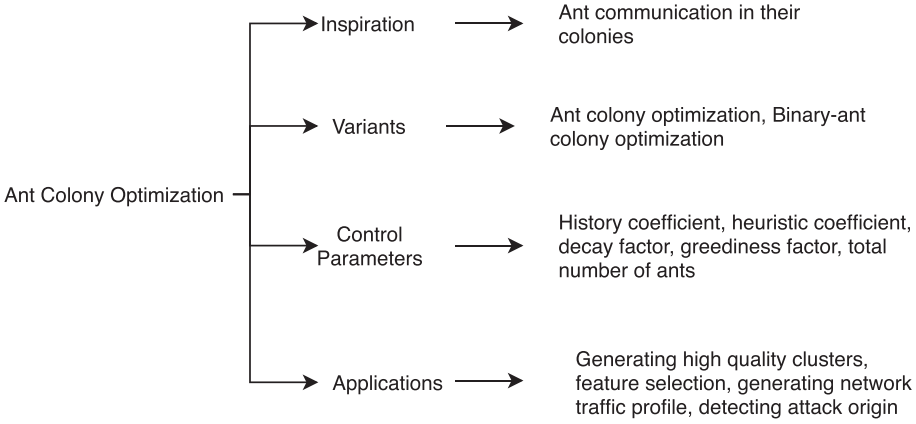


Fig. 6. Ant colony optimization.

features depend on the pheromone trail decay coefficient, pheromone amount, and pheromone evaporation rate. The appropriateness of the chosen parameters defines the quality of the features selected. For instance, if the evaporation is assumed to be zero then the algorithm does not converge, and if the evaporation rate is kept high then algorithm converges to sub-optimal solutions [97]. Therefore, for using ACO as a feature selection method, it should be combined with other feature selection methods for ease of implementation [99]. In Ref. [100], a binary-coded ACO grouped with GA for feature selection. A multi-orientation search process executed using the pheromone density and visibility density model for selecting features. The proposed approach uses SVM classifier for attack classification. The proposed approach represents the features like a fully connected graph and implements two processes, one for selecting the feature and other for discarding. The main aim of feature selection is to achieve high classification accuracy. A fitness function given in equation (6) is used for achieving high classification accuracy. Here, $F(i)$ is the fitness value for ant i, $Accuracy(i)$ is the classification accuracy for ant i, $n(i)$ is the number of features, and λ is the weighting parameter with the value 0.01. The proposed approach gives high classification accuracy and computational efficiency.

$$F(i) = \frac{Accuracy(i)}{1 + \lambda n(i)} \tag{6}$$

Ant colony system can also be used for tracing the origin of the attack which is often referred to as IP traceback problem. To address the issue of tracing the correct routing information, a modified ant colony system is presented in Ref. [101] where a full attack route is traced without having knowledge regarding complete routing information. To achieve the objective, an ant colony with multiple subgroups considered with their own pheromone updation rule. The experimental results generated with ns2 simulations by constructing a network using the Waxman model [101]. The experimentation of the network topology aimed at modeling the ant colony parameters and network attributes. The results showed a slower convergence with a globally optimal solution for detecting the attack route.

ACO is also used for managing the network proactively by identifying the network traffic patterns for classifying normal and abnormal traffic. In Ref. [102], ACO is used for generating the digital signature of the segments flowing in the network. These signatures studied and compared with the stored signatures of the network traffic for IP flow analysis. The attributes considered for flow analysis are the characterization of bits, packets, flow transmitted per second, and retrieving the descriptive flow attributes. The experimental results demonstrated a promising detection of anomalous behaviour in the network with a low FAR. Table 5 summarizes the role of ACO in the field of IDS.

The advantages of ACO-based algorithm are summarized as follows.

- The objective of converging to an optimal solution in ACO is accomplished by continuously updating the pheromone trail. This property of ACO incorporates a positive feedback mechanism in the developed system [84].
- In ACO, a group of ants work in coordination with each other synchronously or asynchronously to achieve optimal solution [84].
- ACO exhibits robustness, as it can be modified for addressing different optimization problems [83].
- ACO algorithm can be hybridized with other heuristic algorithms to enhance the performance of developed model [85].

The disadvantages of ACO are summarized as follows.

- The convergence rate of ACO to an optimal solution is slow compared to other heuristic-based algorithms [84].
- It does not have any centralized controlling entity for directing it towards optimal solutions [87].
- The convergence of ACO algorithm is definite, but the time required for convergence is uncertain [85].
- Moreover, ACO does not perform well with the problems having a large search space [85].

3.3. Particle Swarm Optimization

PSO algorithm is a global optimization algorithm based on the flocking behaviour of birds or schooling behaviour of fishes where the group of agents tries to converge to a common goal by observing the feedback from the other members of the swarm [103]. Here, swarm refers to a large number of decentralized agents with similar characteristics that interact with the other members to achieve optimal solution. This algorithm is often used for the problem domain that is non-linear, discontinuous, and non-differentiable [104]. An optimal solution is derived iteratively based on behaviour of members in the swarm [105]. Every member of the swarm finds a probable solution at the given time based on the interaction between the candidate solution and the other members. Hence, the quality of the solution depends on the candidate solution and the interaction between the members [106]. A variation of PSO, inspired by the inertial influence of the agents is proposed in Ref. [107]. Here, agents are directed toward better solutions based on the present solutions and the varied learning methods [107]. The members move by observing the local optimality and move towards a new candidate solution is often referred to as Levy flights [103]. Here, the members move immediately towards finding solutions for convergence [108]. The PSO algorithm was enhanced by adding the inertia weight factors to effectively handle the exploration and exploitation activities. The attributes of PSO is shown in Fig. 7.

Table 5
Ant colony optimization algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[85]	Parameter Optimization	Not specified	DT-ACO	KDD CUP 99	10% of KDD CUP 99	• DR: 95.19% • FAR: 1.21%	• The proposed method work well with unlabelled data. • The technique is capable of detecting previously unknown attacks.
[86]	Rule generation	Initial pheromone value τ: 0.004, a: 41, b: 6, heuristic probability η: 0.5	Fuzzy Rule Mining-ACO	KDD99	Training set: 752 data samples, Test set: 311,029 data samples	• Classification Accuracy: 94.33% • Classification Cost: 0.1675	• The hybrid method addresses the challenging part of generating rules and tuning the selected rule for further learning. • The idea of addressing multi objective problems to extract classifier rules for classification is yet to be explored.
[88]	Packet filtering	Variable pheromone trail values and energy value	Packet Filtering Algorithm-ACO	Stimulated Dataset	Test set: 40 instances	Comparative analysis with other packet filtering methods is presented.	• The proposed method classifies the packet using multidimensional filters. • ACO based filtering strictly conforms to the rules defined in the firewall and scales well with storage and filtering time.
[93]	Clustering in feature space	Not specified	Kernal based Clustering-KPCA-ACO	Stimulated Dataset	Not mentioned	Results are presented in the form of projections and histograms for analysis.	• The ACO method improves time cost but not clustering quality. • The clustering quality is improved by using KPCA method with ACO.
[95]	Construction of training set	Number of ants: 10, random threshold value: [0,1]	SVM-ACO	KDD99	Training set: 550, Test set: Not mentioned	• Accuracy: 98.62% • MCC: 0.861,161.	• The proposed approach of selecting an appropriate training dataset is not adaptive to multi-classification problems. • However, the gradual feature removal method selects the features that enhances the performance of the classfier.
[96]	Clustering	Swarm similarity coefficient β: 0.25	SVM-ACO	KDD99	Training set: 390 data samples, Test set: 53,604 data samples	• DR: 80.100% • FPR: 2.8% • FNR: 0.360%	• The proposed method implements the idea of clustering combined with active learning to reduce the dimension of the training dataset. • The proposed IDS model is capable of handling binary as well as multi-class classification.
[97]	Feature selection	Initial pheromone, transition probability	SVM-ACO	KDD CUP 99	10% of KDD CUP 99	• CR: 84.27% • Mis-classification rate: 15.7% • Time: 4540.98 s	• The experimental results showed that selecting features using ACO improves the performance of classifier in terms of training time.
[98]	Feature Selection	$\alpha = 0.4$, $\beta = 0.6$, $\rho = 0.2$, the initial pheromone intensity of each feature is equal to 1 ($\tau = 1$) the number of ant in every iteration is 50 (m = 50) and the maximum number of iterations is 100 (T = 100)	ACO	KDD CUP 99	Training set: 494,021, Test set: 2 million	• Accuracy: 98.9% • FPR: 2.59%	• The search capability of ACO is powerful in terms of exploring and selecting the features for classification. • The processing time of ACO is affected by the dimension of the problem. • The proposed method is computationally inexpensive in terms of memory and execution.

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Table 5 *(continued)*

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[102]	Traffic profile generation	Pheromone update	PCA, ACO	Real-time network traffic	7000 nodes	ROC analysis of the real-time network traffic is presented.	• The major contribution of the proposed method is to analyze the qualitative and descriptive attributes of the IP flow. • The proposed method detects the network anomalies and simultaneously provide availability and reliability to the networks.
[99]	Feature selection	σ: 1, γ: 0.01, ρ: 0.9, q0: 0.1, initial pheromone: 0.5, heuristic restricting constant: 0.001	Fuzzy Rough Sets	KDD CUP 99	8368 instances	• Accuracy: 99.81% • Time taken: 1.21s	• ACO-based feature selection resulted in reduction in training, testing and classification time along with feature optimization.
[100]	Feature selection	Number of ants: 20, α: 1.0, β: 0.0–0.3, evaporation of pheromone: [0,1]	BACO, GA	2300 instances	Spambase UCI datasets	Results are presented in the form of evolutionary fitness curve.	• Feature selection using ACO, improves the computational efficiency of SVM for high dimensional datasets.
[101]	For investigating the probable origin of attack	Variable values of control parameters: weighting factor of pheromone intensity, α; the weighting factor of visibility, β; and pheromone decay rate, ρ	Random nodes (100–500)	ACO	Simulated Dataset	Results are presented in the form of graphs and coverage percentage is calculated for the routing packets.	• The proposed method uses routing information for constructing an attack path and investigate the source of an attack.
[89]	Feature selection	Not specified	ACO-SVM	KDD99	Training set: 494,020, Test set: 311,029	• Binary Classification Accuracy: 98.70% • Multi-class Classification Accuracy: 98.73%	• Feature selection using ACO improves the classification accuracy by optimizing the features.

In IDS, PSO is often used to address the issues such as local convergence, high false alarm rates, and selecting prominent features from the feature set. Fuzzy logic and PSO are combined in Ref. [109] to enhance the performance of the fuzzy k-means algorithm. Here, in the proposed method fuzzy inference is used for calculating the fitness values of the fuzzy rules. The fuzzy inference function is implemented using input variable membership function and input variable distance. The objective of the inference process is to differentiate rules as high and low based on their fitness values. However, the proposed method suffers from a high false alarm rate but is computationally inexpensive [109]. In contradiction, when PSO is combined with a traditional k-means clustering algorithm, the grouping of the data samples does not depend on the initialization of the cluster center [110]. This results in better quality clusters when PSO is used for selecting the cluster centers. The experiments are performed on KDD CUP 99 dataset that showed high DR and low FAR. The results also showed improved performance of the k-means clustering algorithm.

A hybrid approach combining PSO and k-means clustering is proposed in Ref. [111], for anomaly detection in content-centric networks. Here, the proposed approach is implemented in two stages: training and detection. In the training stage, clusters are formed by considering Davies Bouldin Index (DBI) and local optimization is performed by considering mean squared error. In the detection stage, combination based distance approaches are implemented such as classification and outlier. The DBI takes into account intra-cluster diversity as well as inter-cluster diversity for defining clusters. The experiments are performed on KDD CUP 99 and results showed that the proposed approach effectively defines the clusters in the training stage and has a high detection rate and low false alarm rate [111].

To increase the accuracy of the IDS model, the idea of selecting relevant features was proposed in Ref. [112] where the standard PSO was applied to optimize the parameters of SVM and binary PSO was used to find prominent features from the feature set. Here, radial basis function was used with the punishment factor C, kernel parameters σ, and the tube size ϵ as user-determined parameters and was optimized using the standard PSO algorithm. Thereafter, binary PSO was applied to the entire feature set by considering each feature as a binary sequence. The detection rate is chosen as the fitness function given by equation (7), where TP denotes the true positive and TN denotes the true negative data instances.

$$Detection_rate = \frac{TP}{TP + TN} * 100 \tag{7}$$

The experiments conducted on KDD CUP 99 and results showed that the proposed approach improved the classification accuracy of the model by selecting free parameters of SVM and features. In the same way, PSO used for fine-tuning the features and combined with a random forest algorithm in Ref. [113] for detecting the probing attack from the NSL-KDD dataset, while in Ref. [114] PSO used for parameter optimization and feature optimization. The supervised learning methods can learn from the already known data and this characteristic is implemented in Ref. [115]. Here, a fast learning network implemented to minimize the number of false positives and false negatives. The performance of the fast learning network optimized by applying PSO. The PSO-based optimization used to optimize the weights and number of neurons in the hidden layer of the fast learning network. The optimization process considers accuracy as the fitness function. The experiments conducted on KDD CUP 99 dataset and the results showed that optimizing the parameters of the fast learning network with PSO improved the testing accuracy for DoS and Probe attack classes. Low testing accuracy was recorded for R2L and U2R attack classes because of the limited amount of training data [115].

Principal Component Analysis (PCA) can be being applied to lower the dimension of a feature set in IDS. The PCA algorithm projects the feature set on a feature space with low dimensions and selects the features having the best eigenvalues. But the features having greater eigenvalues do not comply to give better sensitivity to the classifier. To meet this requirement, feature optimization is required which is performed using PSO with neural networks [116]. A novel approach is presented in Ref. [117] by hybridizing kernel-based PCA with improving chaotic PSO. Here, SVM with RBF has been implemented to reduce the computational time by removing the noise caused due to the feature differences. The improve chaotic PSO is used to optimize the parameters of SVM and kernel-based PCA is used for feature reduction. The parameter optimization is performed using a fitness function shown in equation (8).

$$Fitness = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\frac{1}{m}\sum_{j=1}^{m}(f(x_{ij}) - y_{ij})^2} \tag{8}$$

Here, n denotes the number of folds for cross-validation, m denotes the number of validation subsets, $f(x_{ij})$ is the calculated value of i data samples with j attributes and y_{ij} is the actual value of the data samples. The parameters of SVM are derived by minimizing the fitness function. The experiments are performed on NSL-KDD dataset. The results showed that the proposed model gave high detection accuracy with the parameter optimization compared to a random selection of parameters in SVM. Moreover, identifying principal feature components using KPCA enhances the performance of the model in terms of training and testing time.

PSO applied for spam detection and for improving the random detector generation of negative selection algorithm [118]. The hybrid model of negative selection algorithm with PSO uses the local outlier factor as its fitness function for generating the detector. The detector generation process stops as soon as the estimated spam coverage is reached. The idea for implementing local outlier factor as fitness function is to search for normal data that is used for training. Thus, any data sample that deviates from a certain measure is regarded as an outlier. The fitness function generates unique features for detecting spam in the defined search space. The fitness function used for detecting outlier is capable of evaluating the distance from one point to another along with calculating the density of local neighbours. The fitness function is given by equation (9).

$$LOF_k(i) = \frac{\left(\frac{\sum s\epsilon N_k(i) lrd(s)}{|N_k(i)|}\right)}{lrd(i)} \tag{9}$$

Here, $N_k(i)$ denotes a set of k-nearest neighbours, $lrd(i)$ is the local reachability density, i is the number of particles, and s is number of data samples. The results showed that using a local outlier factor for spam detection improved the performance of the negative selection algorithm.

A hybrid model of IDS is built by combining PSO with random forest classifier in Ref. [119], wherein weighted random forest algorithm is used to extract important and relevant features from the KDD CUP 99 dataset. The obtained candidate feature set then used for classification and PSO used to enhance the performance of random forest algorithm. Here, the PSO algorithm is used for selecting the parameters of random forest such as the number of features to be selected, the number of decision trees, and the dimension of the input training dataset. The most vital part of the proposed approach is to maintain the goodness feature set. Hence, the proposed approach results in high computation time and storage, but the final selected feature set results in a high true positive rate and low false-positive rate. Table 6 summarizes the role of PSO in the field of IDS.

The advantages of PSO for intrusion detection system can be summarized as follows.

- PSO algorithm is a computational technique that iteratively optimizes the problem by improving the solutions based on the given measure of quality [105].
- The PSO algorithm does not need to calculate any crossover and mutation probabilities. Also, the search process is performed based on the speed of the particle. This implies that for every iteration, the

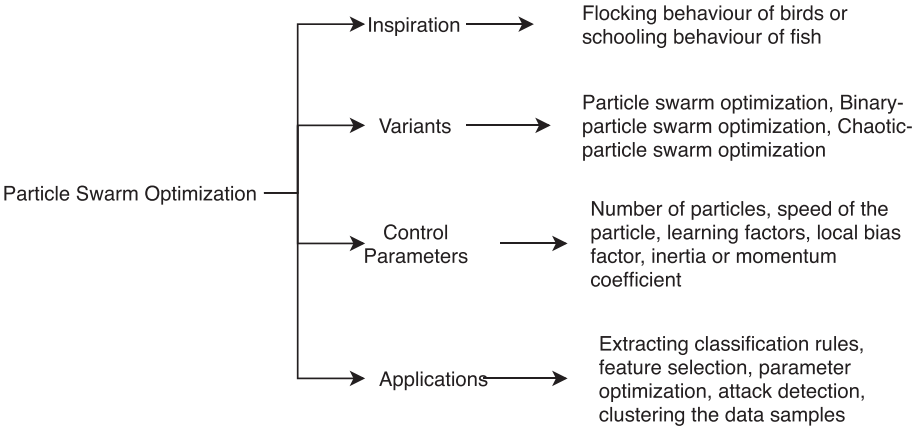


Fig. 7. Particle swarm optimization.

most optimal particle forwards the information to other particles in the swarm. Also, the speed of searching for the optimal particle is fast [105].

- PSO is a meta-heuristic algorithm that takes few or no assumptions for optimizing the given problem. Hence, the evaluations of PSO are simple compared to other algorithms [104].
- Unlike other optimization methods such as gradient descent and quasi-newton techniques, PSO does not require gradient for optimizing problems. This is because PSO considers the problem to be optimized as differentiable [104].
- PSO algorithm can be hybridized with other algorithms to improve the performance of the developed model [106].

The disadvantages of PSO algorithm are summarized as follows.

- The performance of PSO algorithm depends on the selection of control parameters and hence, an optimal solution is not guaranteed [107].
- The behaviour of the swarm in PSO varies with high-dimensional search space and type of the underlying optimization problem such as discontinuous, noise, and time-varying. Thus, this affects the performance of PSO [107].
- A balance between exploration and exploitation is needed to avoid premature convergence and to ensure a good rate of convergence [105].

3.4. Artificial bee colony

ABC algorithm is the optimization method inspired by the bees seeking for their food. ABC looks for the optimal solution from the given number of solutions present [120–122]. The bees are categorized as an employed bee, onlooker bee, and the scout bee. These bees collectively interact with each other to perform tasks like allocating work, food source selection, reproduction, navigating to optimal food sources, and informing other members about the optimal food source by performing waggle dances. Initially, food sources identified from the group of solutions available as the population. Thereafter, the employed bees search for the new food sources which are better than the food sources assigned to them by performing the random search. If the identified solution has a better fitness value than the available solution, it is accepted as the new candidate solution else it is rejected. The employed bees forward this information to onlooker bee by performing waggle dances. The onlooker bees select food source by performing a probabilistic search. If the fitness value of the new solution is better than the available solution, then it is accepted as the new candidate solution for the onlooker bees else it is rejected. If the bees are not able to boost the fitness value of the food source and are exhausted then these bees are abandoned and replaced by the scout bees. The variants of the algorithm studied the characteristics of the queen bee behaviour [123], interactions, and waggle dance [122,124], mating and reproduction characteristics [125]. The attributes of ABC is shown in Fig. 8.

The characteristics of the ABC algorithm can be utilized in IDS by combining it with the learning algorithm considered for the classification. In the study [126], the standard ABC algorithm is used to select the parameters of the SVM classifier whereas binary ABC is used for obtaining optimum feature set. Here, the accuracy rate is chosen as the fitness function. The fitness function reflects the classification accuracy of each feature set and selects the feature set with high accuracy. The experiments conducted on the KDD CUP 99 dataset [94] and the results are compared with PSO and GA combined with SVM. The results showed that hybridization of ABC with SVM yields higher accuracy rate and low FAR.

In Ref. [51], bees algorithm is proposed wherein the set of features is presented as a string of binary sequence. The proposed method applies a random flip mutation operation to execute a random search process in the neighborhood to find the optimal set of features. The SVM classifier is used with an optimal feature set obtained for attack classification. Every feature subset is denoted by a bee and the fitness value of each bee is determined by equation (10).

$$fitness = \alpha * \gamma R(D) + \beta * \frac{|C| - |R|}{|C|} \tag{10}$$

Here, $\gamma R(D)$ is the average of classification accuracy, achieved by performing 10-fold cross-validation. $|R|$ is the length of the feature and $|C|$ is the total number of features. α and β are the two constants that signify the relative importance of accuracy and feature subset length. The value of α ranges between 0 and 1 and value of β is determined by $(1 - \alpha)$. The experiments were performed using KDD CUP 99 dataset for anomaly detection. The optimization implemented using ABC and SVM resulted in improved classification accuracy and false alarm rate [51].

A functional architecture for network-based IDS is defined in Ref. [127] where ABC is used for defining the classification rule for recognizing patterns from the network traffic. For defining the classification rule, the lower bound and upper bound value of each feature is evaluated based on its maximum and minimum values, which is further used for computing the fitness value of the feature. Features with the highest fitness value are selected for detecting intrusions. The proposed approach uses the standard accuracy rate as the fitness function. ABC is used for optimizing the parameters of SVM and information gain is used for extracting features in Ref. [128]. The experiments are performed on NSL-KDD dataset and results are compared with the standard SVM. The proposed framework combining ABC and SVM gives a high DR and low FAR.

Table 6
Particle swarm optimization algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[109]	Parameter Optimization	Not specified	Fuzzy k-means-PSO	KDD CUP 99	Training set: 98,095 data samples, Test set: 311,028 data samples	• DR: 95.87% • FAR: 2.12%	• The proposed approach is inexpensive in terms of CPU time and memory. • The hybridization with PSO, solves the local convergence problem of k-means algorithm. • The proposed algorithm recorded high false alarm rate.
[112]	Parameter optimization and feature selection	learning factors c1 and c2: 2.3 and 1.8, inertia weight w: 0.9 to 0.5, maximum fly velocity: 200	SVM-PSO	KDD CUP 99	Not mentioned	• DR: 99.84%	• The proposed approach selected the parameters for SVM as well as features for improving the performance of the classifier.
[110]	Parameter optimization	learning factors c1 and c2: 2, inertia weight w: 0.9 to 0.4, maximum fly velocity: 200	k-means, PSO	KDD CUP 99	Training set: 97,278 data samples, Test set: 301,129 data samples	• DR: 74.43% • FAR: 1.86%	• Solves the problem of k-means algorithm, of falling to local optima. • Faster processing time and convergence rate.
[113]	Feature selection	learning factors c1 and c2: 2, inertia weight w: 0.9 to 0.4, maximum fly velocity: 200	RF-BPSO	KDD CUP 99	494,021 instances	• DR: 99.92% • FAR: 0.029%	• The optimization of selected features with PSO achieves the highest average detection rate and the lowest false positive rate.
[114]	Parameter optimization and feature optimization	Not specified	SVM-PSO	NSL-KDD	4900000 instances	•Accuracy: 81.8%	• The parameters of SVM are selected using PSO. Also, feature selection is performed by PSO. • The proposed algorithm is expensive in terms of computational time and memory usage.
[116]	Feature selection	learning factors c1 and c2: 2, inertia weight w: 0.9 to 0.4	NN-PSO	Simulated Dataset	Training set: 3750 data samples, Test set: 2250 data samples	• DR: 99.4% • FAR: 0.6%	• The proposed techniques applies PCA for transforming the features and PSO for selecting the features from the principal components in the search space. • The training time required for NN is also reduced by using the hybridization of PCA and PSO.
[117]	Parameter Optimization	Cmax: 300, Wmax: 0.9 Wmin: 0.4, Cmax: 2.0, Cmin: 1.4, H: 1.	SVM-PCA-Imporved chaotic PSO	KDD CUP 99	60,254 instances	• DR: 96.372% • FAR: 0.97% •CC: 0.97	• The implementation of feature extraction using KPCA improves the generalization performance of SVM. • The proposed method is implemented for binary classification. • The objective of hybridizing KPCA and improved chaotic PSO with SVM is to improvise not only detection rate and false alarm rate but also testing and training time.

(*continued on next page*)

Table 6 *(continued)*

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[111]	Parameter optimization	W start: 0.9, W end: 0.4, and c1 = c2 = 2.	PSO-Kmeans	KDD CUP 99	5770 instances	• ROC curves for the proposed method is presented • AUC: 97.4% • Mean Accuracy: 94.4% • Mean SD: 0.97 • Mean Recall: 96.8% • Mean Specificity: 95.52%	• The proposed method improves the performance of the algorithm in terms of computational time and accuracy. • However, non-linear functions and multi-objective optimization are not addressed.
[118]	Parameter optimization	Not specified	PSO	email spam dataset	4601 messages	• Accuracy: 82.62% • CC: 63.37% • F1: 74.94% • Sensitivity:65.99% • Specificity:93.42% • PPV: 86.71% • NPV: 80.87%	• The advantage of proposed techniques is that, it identifies spam messages based on existing patterns.
[119]	Parameter optimization	Not Specified	PSO-RF	KDD CUP 99	Not mentioned	Trade-off between the TPR and FPR is presented.	• Here, the feature selection process is enhanced by using RF with PSO and high attack detection and classification rate is achieved.
[115]	Weight optimization in FLN	c1 = c2 = 1.42, w = 0.75 and number of particles = 50	FLN-PSO	KDD CUP 99	4900000 instances	Testing accuracy curve with respect to different neuron values is shown along with a confusion matrix table for all attack categories.	The weight optimization of FLN by PSO improves the performance of the classifier in terms of accuracy. • Also, with increase in the number of neurons the accuracy gradually improves.

In Ref. [129], ABC is combined with a modified NB algorithm where ABC is used to optimize the weights of the modified NB algorithm. Here, in the proposed method, weights are assigned to the features. The features having a higher impact on the results are assigned higher weights and features having a lower impact on the results are assigned lower weights. The weights are real numbers in the range of 0–2 [129]. The quality of the features is governed by the fitness function. The classification accuracy of modified NB is considered as the fitness function denoted by $f(i)$ and is given by equation (11). Here, *True* is the correctly identifies data samples and *False* is incorrectly identified data samples. The experiments are tested on NSL-KDD dataset that showed that ABC is capable of handling high-dimensional datasets.

$$f(i) = \frac{True}{True + False} \tag{11}$$

Hybridization of ABC with the learning algorithm has also enhanced the efficiency and effectiveness of the IDS model. ABC is hybridized with ANN in Ref. [130] to optimize the weights and biases of the connecting edges in the network. Here, the weights and biases are optimized using mean squared error as the fitness function that is given by equation (12).

$$MSE = \frac{1}{2}\sum_{i=1}^{m}(T_i - O_i)^2 \tag{12}$$

Here, T_i denotes the actual output value while O_i denotes the estimated output value and m is the total number of instances. The experiments were conducted on NSL-KDD dataset and results were shown in terms of accuracy and error rate. The optimization using ABC improved the learning ability of the neural network. This is because the best weights and bias are chosen that minimizes the mean squared error.

A multi-objective ABC is proposed in Ref. [131] with an aim to eliminate redundant features from the feature set and is also hybridized with PSO to optimize the parameters of feed-forward NN. The features selected through an evolutionary selection procedure. The fitness function is defined that shows a trade-off between the classification error rate, false alarm rate, and selected features. These are the three objectives to achieve high classification accuracy. The fitness function is given by equation (13).

$$fitness = \alpha * \left(\frac{SelectedFeatures}{AllFeatures}\right) + \beta * \left(\frac{FalseRate}{AllFalse}\right) + \beta * \left(\frac{ErrorRate}{AllError}\right) \tag{13}$$

Here, α and β are the constants signifying the relative importance of selected features, false rate, and error rate. The value for α is between 0 and 1 and value of β is evaluated as $\frac{(1-\alpha)}{2}$. The *AllFalse* and *AllError* are the false alarm and error rate of all the available features, respectively. Thus, the proposed feature selection method represents multi-objective ABC as a group of non-dominated solutions that are used as feature subset [131].

To improve the performance of IDS, a hybrid classification model is developed using ABC and artificial fish swarm optimization [132]. In the proposed method fuzzy c-means clustering and correlation-based feature selection method are used for preprocessing the dataset, and classification rule is generated by classification and regression tree. The hybrid classification model is then trained using the classification rule. The experiments performed on NSL-KDD and UNSW-NB15 datasets. The results showed DR of approximately 99% with 0.01% FPR. Table 7 summarizes the role of ABC in the field of IDS.

The advantages of ABC are as follows.

- ABC algorithm is easy to implement, robust, and highly flexible [120].
- Addition or reduction in the number of bees does not require re-initialization of the algorithm. This is because the performance of the algorithm is de.

Pendent on only two control parameters, maximum cycle number and colony size [120].

- ABC algorithm can be hybridized with other algorithms to address various optimization problems. Also, it requires very fewer parameters compared to other searching techniques [121].

The disadvantages of ABC are as follows.

- In order to improve the performance of the model, control parameters need to be modified that require new fitness tests [122].
- The convergence rate to obtain optimal solution is slow in case of serial processing [124].
- ABC requires high amount of objective function evaluation to achieve optimal solution [124].

3.5. Firefly algorithm

Firefly Algorithm (FA) inspired by the lightning of the firefly species to solve complex problems having high variance and complicated objective functions with constraints [133–136]. Compared to the other swarm and evolutionary algorithms firefly algorithm can be applied to multi-modal functions very efficiently [137]. This algorithm is a population-based search approach and therefore, the new solutions have an advantage over solutions obtained so far. It efficiently balances exploration and exploitation activities by parameter tuning. Every firefly species is having a bioluminescence property which emits light through the solution called luciferin. This characteristic helps the firefly to attract other fireflies for mating or protecting themselves from their enemies. The FA is a self-organized and decentralized algorithm. The attractiveness of the firefly is dependent upon the brightness of the light that it emits. The firefly with low brightness is attracted to the firefly with greater brightness.

To define the standards of the FA, it is assumed that all the fireflies are unisex and the attractiveness of the firefly is defined by the objective function which is defined by the light intensity [135]. The objective function denotes the fitness value of the candidate solution. The population of the firefly is initialized randomly and the fireflies are evaluated based on the objective function and are ranked according to the highest value of the fitness. This procedure is repeated until a stopping criterion is met which usually is the number of generations defined in advance. The FA can be combined with other algorithms to improve the efficiency [138–140]. The attributes of FA is shown in Fig. 9.

The applicability of FA in the field of IDS can be described in terms of optimizing the intrusion detection process by enhancing the efficiency of the classifier as well as an optimizer for the feature selection process. For instance, in Ref. [141], FA is combined with k-harmonic means algorithm to enhance the clustering process by minimizing the cluster initialization problem to detect network anomalies. The objective of using FA along with k-harmonic means algorithm is to optimize the initialization of k-clusters. The optimization function used for this objective is shown in equation (14).

$$KHM(X, C) = \sum_{i=1}^{n} \frac{k}{\sum_{j=1}^{k} \frac{1}{\|x_i - c_j\|^p}} \tag{14}$$

Here, $X = x_1, x_2, \ldots, x_n$ is the data to be clustered, $C = c_1, c_2, \ldots, c_k$ is the set of cluster centers. $\| x_i - c_j \|^p$ is the membership function determining the relationship strength of data x_i with respect to center c_j and p is the input parameter that is assumned to be greater than or equal to 2 [141]. Applying this hybrid method results in effectively escaping the local optima solutions by creating clusters of network data for identifying anomalies.

An adaptive Snort plug-in model is proposed in Ref. [142] that executes the snort rule set in parallel. The proposed framework is capable of handling both known and unknown malicious traffic. The proposed plug-in consists of different modules for decoding the network pack-

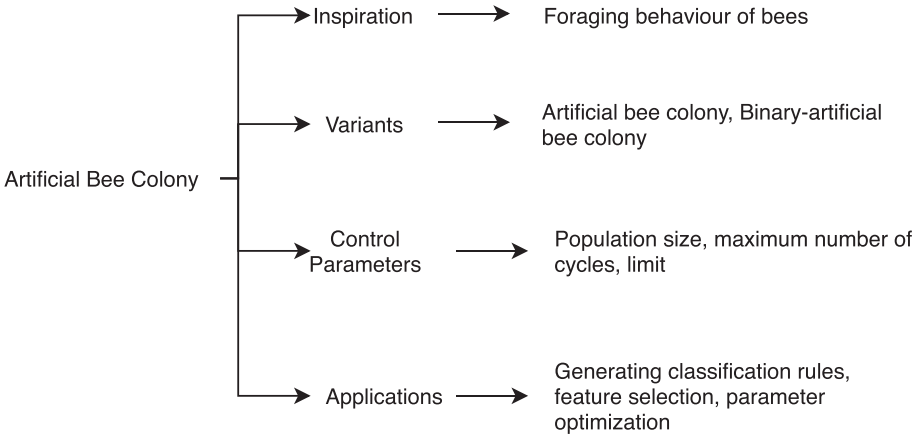


Fig. 8. Artificial bee colony.

ets and classifying different malicious activities found in the network. Classifiers such as DT, fuzzy logic, and different variants of SVM are used for building the hybrid model. For instance, a hybrid model of SVM and fuzzy logic and SVM optimized with FA are implemented. The experiments conducted on the snort dataset and results showed that SVM optimized with FA outperforms all the other algorithms. An optimized hybrid model is proposed in Ref. [143] wherein a rule-based model with self-organizing maps is combined with FA to differentiate the network traffic activity without any dependence on payloads or ports. The proposed model forms association rules for classification and the parameters of the rule-based model as well as self-organizing maps are optimized using the FA algorithm. A metaheuristic based detector generation is studied in Ref. [144]. Here, FA is hybridized with GA to find the centroid of the network traffic samples. The proposed framework generates detectors for network anomaly detection. The process of generating detectors is controlled by an objective function given by equation (15).

$$f(s_i) = \begin{cases} N_{abnormal}(s_i) - N_{normal}(s_i), & \text{itr} = 1 \\ N_{abnormal}(s_i) - N_{normal}(s_i) + old_intersect(s_i), & \text{itr} > 1 \end{cases} \quad (15)$$

Where *itr* is a number of iterations for detector generation. $N_{normal}(s_i)$ is the number of normal data samples identified by the detector, $N_{abnormal}(s_i)$ is the number of abnormal data sample identified by the detector, $old_intersect(s_i)$ is the percentage of normal samples that are identified by the detector in the previous generations and s_i is the number of data samples. Thus, selecting a suitable number of detectors resulted in effective online processing of data samples.

A hybrid FA is proposed in Ref. [145] where FA is hybridized with simulated annealing to enhance the search process and classification. The hybrid FA selects the best firefly by calculating the intensity of each firefly using equation (16).

$$Intensity_i = \frac{1}{\sqrt{\sum_{j=1}^{attr} (X_{test,j} - X_{i,j})^2}} \quad (16)$$

Here, $X_{test,j}$ signifies the *j*th attribute of the test data and $X_{i,j}$ signifies *j*th attribute of firefly *i*. The fireflies with best intensity values are chosen and given as an input to simulated annealing for deriving the best optimal solution. With this approach, the results are obtained for KDD CUP 99, NSL-KDD, and UNSW [146] datasets and compared with tree classifiers such as J48, NB tree, NB, MLP, and RFT. The result analysis showed that the hybrid FA outperforms all the techniques for detecting anomalies in discreet search spaces.

FA can also be implemented as a wrapper-based method to optimize the feature selection process. For instance, in Ref. [147] FA is used to find optimal feature set and binary FA is used as a feature selection technique in Ref. [148], where all the fireflies are initialized randomly and all the features are considered as binary sequences with the series of zeros and ones. Here, both methods use the fitness function given in equation (17).

$$Error = \sigma * \frac{\#Features}{\#AllFeatures} + (1 - \sigma) * \frac{Err[\#Features]}{Err[\#AllFeatures]} \quad (17)$$

The objective of the fitness function is to reduce the error rate of the classifier for the training and validation set. The experiments conducted on NSL-KDD dataset and the comparative analysis shown for other approaches. Here, *#Features* are the selected features from the dataset, *#AllFeatures* is the entire feature set, *Err* is the error rate of the classifier, and σ is a constant value between 0 and 1.

In Ref. [149], FA is used with C4.5 and NB as a feature selection approach. A mutual adaptive strategy is applied for selecting the features from the feature set. Hence, features with highest mutual information are selected by updating their postion using the rule defined in equation (18).

$$v_{id}^{t+1} = \begin{cases} 1, & if\ p_{id} > rand \\ 0, & otherwise \end{cases} \quad (18)$$

Here, v_{id}^{t+1} is the current value of *d*th feature in *i*th firefly, p_{id} represents previous value of *d*th feature in *i*th firefly, *t* is the number of iterations, and rand is the random number between 0 and 1. The experiments performed on KDD CUP 99 dataset and the results showed that only 10 features are sufficient enough to improve the accuracy of the system. Table 8 summarizes the role of FA in the field of IDS. FA has following advantages.

- FA is based on attractiveness between the fireflies and therefore, these fireflies in the swarm can divide themselves into subgroups. The subgroups can move around each other or local optima to achieve the best global solution [150].
- The division into subgroups enables the fireflies to find optimal solution simultaneously even for the larger population size [133].

FA has also shown a few drawbacks which are as follows.

- FA quite often gets trapped into local optima as they perform a local search to find the best solution [134].
- The control parameters of FA do not vary with time and are static [134].
- FA has a disadvantage of not memorizing any previous solution for each firefly present in the population. This results in fireflies moving in the search space regardless of the previous better solution and may result in completely missing the best solution [137].

Table 7
Artificial bee colony algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[126]	Parameter optimization and feature selection	Colony size: 20, maximum evaluation number: 100, limit: 50	SVM-ABC	KDD CUP 99	Training set: 6092 data samples, Test set: 5890 data samples	• Accuracy: 99.92%	• The ABC algorithm not only selects features but also improves the performance of SVM by selecting the best appropriate parameters. • The technique requires more time to select the parameters and feature set for evaluation.
[51]	Feature selection	colony size: 25, Number of recruited bees for best m sites: 15, Number of recruited bees for the best e sites: 20	SVM-ABC	KDD CUP 99	Training set: 4000, Test set: 12,000	• DR: 95.75% • FAR: 0.16%	• The feature selection performed using ABC, produced improved results with different lengths of feature sets.
[127]	Rule extraction	Variable population size, maximum generation number, and limit	ABC	KDD CUP 99	Training set: Not mentioned, Test set: 4000 instances	• Average Accuracy:98.5% • Best Accuracy:99.5% • SD: 0.004	• The proposed method showed that the accuracy of the system remain steady with increase in the maximum generation number. • Also, as the limit parameter is increased, the accuracy of the model did not change significantly. • Thus, the proposed method is able to retain the robustness of the system with variable population sizes.
[128]	Parameter optimization	Colony size: 20, limit: 40, and problem dimension: 2	SVM-ABC-IG	NSL-KDD	9566 instances	• Accuracy: 98.89% • DR: 98.53% • FAR: 0.037	• The parameter selection using ABC and feature selection using IG improved the performance of SVM. • The execution time recorded is also less compared to traditional SVM algorithm.
[130]	Weight optimization	Variable population size, maximum generation number, and limit	ANN-ABC	NSL-KDD	Training set: 125,973 data samples, Test set: 22,544 data samples	• Accuracy: 87.27% • Error rate: 0.126%	• The accuracy obtained by using ABC for optimizing weights of MLP is less as the performance was tested using all the features of the dataset.
[131]	Feature selection and optimization	Variable population size, maximum generation number, and limit	ABC	KDD CUP 99	Not mentioned	The method to be implemented is discussed in detail.	• A multi-objective ABC is presented to optimize the feature set for improving the classification error rate as well as classifier performance.
[129]	Parameter optimization	Random number: [0,1], population size: 60, proportion of employment bees: 0.5	ABC-MNB	NSL-KDD	Training set: 5000 Test set: 5000	• Accuracy: 91.08% • Sensitivity: 0.86% • Specificity: 0.94% • F-score: 0.90%	• The strong convergence capability of ABC algorithm, fastens the processing of MNB algorithm. • Also, scout bees prevent the model to converge to local optima and thus, giving diversity of solutions.
[132]	Attack classification	Population size: 30, limit: 5000, problem dimension:1500	ABC-AFS	NSL-KDD, UNSW-NB15	For NSL-KDD: Training set is 125,973 records and Test set is 22,544 records and for UNSW-NB15: Approximately 3 million records	• Accuracy: 99.0% • FPR: 0.01%	• The evaluation of proposed method in terms of computation cost and time is less considering the diversity of the network data.

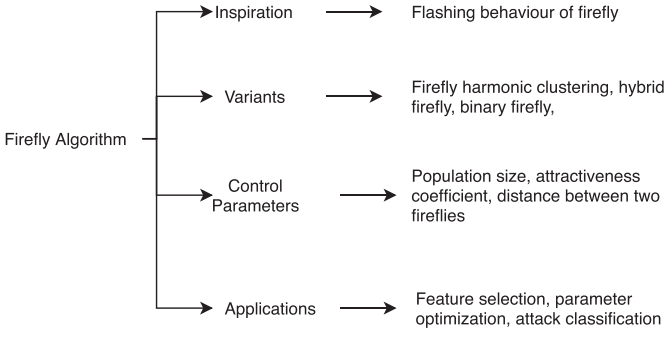


Fig. 9. Firefly algorithm.

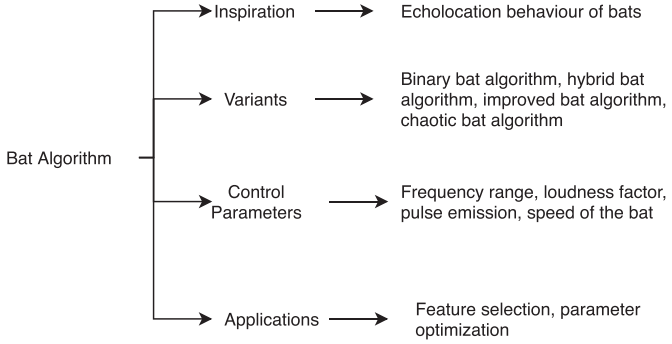


Fig. 10. Bat algorithm.

3.6. Bat algorithm

Bat Algorithm (BA) is inspired by the echolocation behaviour of bats and is used to solve single-objective and multi-objective optimization problems. The echolocation behaviour of bats can be defined as the technique based on the sound of echoes generated by the bats that they use to move in their surrounding environment. This helps the bats to locate their prey in darkness. Echolocation is used to estimate the distance between food and objects.

The bats can modulate their flying velocity as well as the frequency and intensity of their voice during the search for prey. The process of searching is carried out iteratively using vector algebra. In each iteration, the frequency and the loudness of the echo emitted by the bat are updated in such a way that as the bats find a potential prey the frequency increases and the loudness decreases. The bat improves its position by updating its personal and global history. The personal and global history of the bats are updated by updated the frequency ($freq_i$), velocity (v_i), and position (x_i) of the bats using equations (19)–(21), respectively.

$$freq_i = freq_{min} + (freq_{max} - freq_{min}) * \beta \tag{19}$$

Here, $freq_i$ denotes the frequency of bat i, $freq_{min}$ and $freq_{max}$ are the minimum and maximum frequency of the bats respectively, β is a constant between 0 and 1.

$$v_{i,j^t} = v_{i,j^{t-1}} + (x_{i,j^{t-1}} - xbest_j) * freq_i \tag{20}$$

Here, v_{i,j^t} is the velocity of bat i at iteration t and $v_{i,j^{t-1}}$ is the velocity of bat i at iteration $t - 1$, x_{i,j^t} is the position of bat i at iteration t and $x_{i,j^{t-1}}$ is the position of bat i at iteration $t - 1$, and $xbest$ is the best position known so far.

$$x_{i,j^t} = x_{i,j^{t-1}} + v_{i,j^t} \tag{21}$$

The bat algorithm is used to address the problems which are continuous in nature while binary bat algorithm used to solve discreet decision-making problem [151]. Another chaotic version of the bat algorithm was proposed for global optimization of the search mobility problem of BA [152]. The hybridization of the bat algorithm has also been studied to improve the efficiency of the SVM algorithm [153]. The attributes of BA is shown in Fig. 10.

To improve the detection accuracy of the IDS model, BA has been used for optimizing the feature selection process, parameters of the classifier, and detecting the intrusions and classifying them into the defined attack classes. BA used to optimize the SVM parameters for intrusion detection [154]. Here, in the proposed work [154], BA is used to optimize the regularization parameter and kernel parameter of SVM. The experiments are tested on NSL-KDD dataset and features are selected using information gain algorithm. A network anomaly-based IDS is proposed in Ref. [151] where SVM is combined with the improved version of BA. Here, the standard BA is used for optimizing the input parameters of the SVM and binary variant of BA is used as the wrapper-based feature selection technique for selecting features from the dataset. In both methods, the attack detection accuracy of the classifier is improved by using the BA algorithm. Here, BA is implemented in two stages that are exploitation and exploration. The exploitation stage is improved by adjusting the loudness and pulse emission rate given by equations (22) and (23), respectively. The exploration stage includes local search and is implemented using equation (24). The experiments are conducted on the NSL-KDD datasets and are compared with standard SVM, PSO and BA based approaches.

$$A_i^{t+1} = \alpha * A_i^t \tag{22}$$

In equation (22), A_i^t is the loudness of bat i at iteration t, A_i^{t+1} is the loudness of bat i at iteration $t + 1$, and α is the constant between 0 and 1.

$$r_i^{t+1} = r_i^0 * [1 - e^{-\gamma * t}] \tag{23}$$

In equation (23), r_i^0 is the initial value of pulse emission rate, r_i^t is the pulse emission rate of bat i at iteration t, r_i^{t+1} is the pluse emission rate of bat i at iteration $t + 1$, and γ is the constant between 0 and 1.

$$x_{new} = x_{old} + \delta * A^*{}_t \tag{24}$$

In equation (24), x_{new} is the new position of the bat, x_{old} is the previous position of bat, δ is a constant between 0 and 1, and A_t^* is the average loudness of all bat at iteration t.

The variants of BA algorithm such as binary BA, improved BA, and hybrid BA have been used for IDS. For instance, an improved version of binary BA with Levy flights used as a wrapper-based feature selection method in Refs. [155,156]. The fitness function used for these proposed feature selection method considers accuracy and number of features. The consideration of accuracy and number of features in the fitness function ensure that the model is capable of generating alerts for identified attacks. For instance, the fitness function used in Refs. [155,156] is given by equations (25) and (26), respectively.

$$fitness = 90\%Accuracy + 10\%\left(\frac{1}{nbFeat}\right) \tag{25}$$

$$fitness = 60\%ADR + 30\%\left(\frac{1}{FAR}\right) + 10\%\left(\frac{1}{nbFeat}\right) \tag{26}$$

Here, $nbFeat$ is number of selected features, ADR is the Attack Detection Rate, and FAR is the False Alarm Rate.

The binary version of BA executes the search process in a multidimensional binary grid by considering the features as the swarm of bats in the form of series of zeros and ones. The binary BA reduces the features by 60%–64% and improves the detection rate and accuracy of the IDS model. The hybrid BA used for optimizing resource consumption in Ref. [153]. The hybrid BA along with SVM is used for feature selection and optimizing parameters of SVM. Table 9 summarizes the role of BA in the field of IDS.

The feature selection process using BA implemented in Refs. [155–157] uses Levy distribution instead of random walks for finding

Table 8
Firefly algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[141]	Parameter Initialization	For harmonic clustering K: 2,3,4,5	FHC	Real Traffic Data	Not mentioned	ROC curve has been presented for comparative analysis of results.	• The proposed algorithm addresses the limitation of initializing cluster centers for the k-means algorithm. • The proposed technique achieved high accuracy compared to k-means because of the variation in the digital signature in the network segments.
[143]	Parameter optimization	Alpha Coefficient α: 0.8, Minimum Attractiveness β0: 1, Absorption γ: 1	SOM, FA	eMule, Skype	1,000,000 for each application	For eMule dataset • Accuracy: 100% • CR: 94.06% • Completeness: 94.06% • FPR: 0% • FNR: 0% For Skype dataset • Accuracy: 100% • CR: 98.06% • Completeness: 98.06% • FPR: 0% • FNR: 0.22%	• The proposed method can easily cope with encrypted data and extract patterns from data collected through application protocols.
[144]	Parameter initialization	value of k: 2,4,6	GA-FA	Real Traffic Data	Training set: 1,074,992, Test set: 77,289	ROC analysis of the results has been presented.	• The hybrid approach is capable of handling large scale datasets. • It maintains the diversity of data samples by selecting the starting points for detector generation. • It requires less time for online anomaly detection as the number of detector generated are reduced.
[145]	Attack detection and classification	Not specified	HFA	KDD CUP 99, NSL-KDD, UNSW	KDD: 494,021, NSL-KDD: 125,973, UNSW: 82,332	• Accuracy: 84.89% • Precision: 84.28% • Recall: 85.50% • F-score: 89.5%	• The hybrid approach performs probabilistic based search for finding the optimal solution. • However, the proposed method records low precision values for attack classification.
[142]	Parameter optimization	Not specified	SVM-FA	Snort and Suricata	1,000,000 TCP, UDP and ICMP packets each	• Accuracy: 95% • FPR: 8.6% • FNR: 2.2%	• The multi-threaded architecture of snort and suricata used for evaluation utilized more memory and CPU time for processing. • The hybrid approach produced improved results for both applications with low false positive and false negative rate.
[148]	Feature selection	$B_{min} = 0.0$, G = 1.0, A = 0.2, D = 0.96	FA-NB	NSL-KDD	Training set: 5 million, Test set: 3 million	• Accuracy: 92.02% • Error rate: 7.98%	• The randomization of FA improves the performance of NB classifier in terms of accuracy and error rate.
[147]	Feature selection	$B_{min} = 0.0$, G = 1.0, A = 0.2, D = 0.96	BFA-NB	NSL-KDD	Training set: 5 million, Test set: 3 million	• Accuracy: 94.83%	• The preprocessing of the data samples and selecting features using BFA improved the accuracy of the system. • The selected features were considered as the sequence of binary strings and hamming distance is used for calculating the distance that improved the movement of the fireflies.
[149]	Feature selection	$\alpha = 0.1$ (randomization parameter), $B_0 = 1$ (Base attraction), $\gamma = 1$ (Absorption coefficient)	C4.5-NB-FA	KDD CUP 99	Training set: 179,116 data samples, Test set: 179,636 data samples	For NB: • Accuracy: 98.9% • FPR: 0.01% • F-measure: 0.99 For C4.5: • Accuracy: 99.5% • FPR: 0.03% • F-measure: 0.97	• Incorporating FA for selecting features improved the performance of classifiers in terms of detection and computational cost.

Table 9
Bat algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[157]	Feature selection	Maximum loudness A0 = 10, minimum pulse rate r0 = 0.9, convergence parameters $\gamma = 0.1$ and $\alpha = 0.9$, range the frequency between 0.8 and 1.0, and $\eta = 0.1$ for Levy flights.	SVM-BBA	NSL-KDD	Training set: 9500. Test set: 4500	• Accuracy: 90.06% • DR: 95.05% • FAR: 4.4%	• The limitation of convergence to local optima of binary BA is overcomed by the proposed method. • The proposed method is only tested with SVM classifier and hence more comparative analysis is needed to range the efficiency of the method.
[153]	Feature selection and parameter selection	Initial loudness A0 = 0.5, initial pulse rate r0 = 0.5 and the frequency interval range from 0.0 to 2.0	SVM-HBA	NSL-KDD	Not mentioned	• Accuracy: 99.28% • FAR: 0.48%	• The proposed method reduces the size of the dataset by 62% and yields better accuracy than SVM.
[151]	Feature selection and parameter optimization	Initial loudness A0 = 0.5, initial pulse rate r0 = 0.5 and the frequency interval range from 0.0 to 2.0	SVM-BA	NSL-KDD	Not mentioned	• ADR: 96.88% • FAR: 1.8%	• The feature selection and parameter optimization using BA, results in effectively streamline the network data features and improve the performance of the classifier.
[155]	Feature selection	Maximum loudness A0: 0.5, minimum pulse rate r0: 0.5, frequency ranges between 0.8 and 1.0 γ: 0.1 and α: 0.9	SVM-NB-BBA	NSL-KDD	Training set: 9566, Test set: 4500	With NB: • ADR: 91.62% • FAR: 5.73% With SVM: • ADR: 95.78% • FAR: 2.89%	• The proposed algorithm improved the performance of NB and SVM classifier in terms of attack detection rate and false alarm rate. However, the results decreases for unknown intrusions.
[154]	Attack detection and classification	Maximum loudness A0: 0.5, minimum pulse rate r0: 0.5, frequency ranges between 0.8 and 1.0 γ: 0.1 and α: 0.9	SVM-BA	NSL-KDD	9566 records	• Accuracy: 99.38% • DR: 99.15% • FAR: 1.9%	• The proposed method improves the performance of SVM classifier in terms of detection accuracy and false alarm rate.
[156]	Feature selection	Maximum loudness A0: 0.5, minimum pulse rate r0: 0.5, frequency ranges between 0.8 and 1.0 γ: 0.1 and α: 0.9	SVM-C4.5-BBA	NSL-KDD	Training set: 9566, Test set: 4500	With C4.5: • ADR: 96.02% • FAR: 2.75% With SVM: • ADR: 96.88% • FAR: 1.8%	• The proposed approach takes comparatively less number of iterations to reduce the features. • The reduced number of features result in making the decision process fast and realiable.

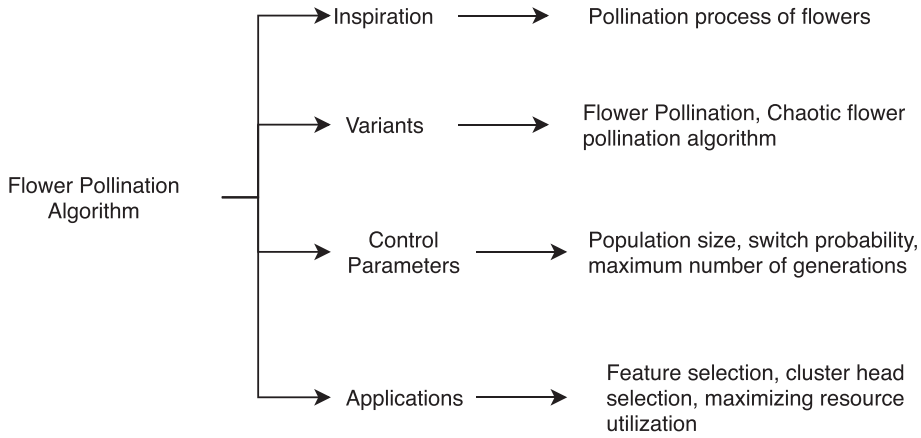


Fig. 11. Flower pollination algorithm.

the best solution and escaping early convergence to local optima. Therefore, equation (24) is modified as shown in equation (27).

$$x_{new} = x_{old} + t^{-\eta} * A_t^* \tag{27}$$

Here, $t^{-\eta}$ defines the Levy distribution and value of η ranges between 1 and 3.

Based on the work reviewed, it is observed that BA is efficient for feature selection. The literature studied showed that the BA can handle the problem of the high dimensionality and finding the most informative features in a search space. This is achieved by replacing the uniform distribution in search space with Levy distribution. BA can be improved further by tuning control parameters during iterative search process to enhance the convergence speed. The advantages of BA are summarized as follows.

- BA is simple, flexible, and easy to implement and works well with complex problems [157].
- BA works well with non-linear problems and can be applied to a wide range of problems [157].
- BA has a quick transaction from exploration to exploitation. This characteristic of BA algorithm allows them to have a fast convergence rate [154].
- The control parameters of the BA algorithm such as loudness and pulse emission rates provide a provision for automatic control and self-optimization. This characteristic of BA gives good optimal solutions [156].
- BA algorithm has the property of auto-zooming into the region where good optimal solutions are present [156].

The disadvantages of BA are summarized as follows.

- The switching of BA is guided from the exploration stage to the exploitation stage. Hence, with varying values of loudness and pulse emission rates may lead to stagnation after some initial point [154].
- There is no mathematical analysis to establish the relation between the parameters and convergence rate [154].
- The accuracy of the BA-based systems depends on the number of function evaluations [153].

3.7. Flower Pollination Algorithm

The Flower Pollination Algorithm (FPA) inspired by the pollination process of a flower; is a global optimization problem with multiple objectives [150]. The pollination process can be defined as the reproduction process where flowers transfer their pollen to other flowers of the same species or flowers of different species to germinate. The transfer of pollen is carried out by pollinators such as insects, wind, water, and animals. The pollination process carried out by insects or animals is referred to as biotic pollination and pollination process carried out by water and wind is referred to as abiotic pollination [158]. The pollination process can be of two types: self-pollination (flowers of the same plant) and cross-pollination (flowers of different plants) [159]. The biotic and cross-pollination is referred to as the global optimization process and abiotic and self-pollination is referred to as the local optimization process. The algorithm is implemented on the assumption that every flower consists of at least one pollen for reproduction. The algorithm aims to find the candidate solution with the highest fitness value. The solution depends on the step size of the consecutive iterations and the pollination process. The step size in global optimization is calculated using the levy flights while random uniform distribution is considered for the local optimization [150]. If the new candidate solutions obtained have greater fitness value than the previous ones then the candidate solutions are updated else the new candidate solutions are eliminated. This process is continued until the required candidate solution is obtained. The algorithm has some challenges in the form of time complexity and multidimensionality as described in Ref. [159]. The attributes of FPA is shown in Fig. 11.

The local and global optimization properties of FPA have been utilized for optimizing feature selection process in IDS. In study [160], the features are selected using the pollination based optimization technique. In the feature selection process, the dataset is considered as plants, the number of data samples in the dataset is the population, and features are considered as the pollinators. The selected features are given as input to two different classifiers, NB and J48. The proposed approach is compared with GA and comparative analysis is presented by considering 20 pollinators.

Likewise, in Ref. [161] feature selection process is optimized by assigning relative weights to each feature and learning these features by building stochastic gradient descent and adaptive boost model. The feature set is also obtained by applying a hybrid feature selection method as proposed in Ref. [162]. Here, the hybridization is performed by combining PSO with FPA to reduce the FAR and to increase the accuracy of the system by selecting the features from the entire feature set. Table 10 summarizes the role of FPA in the field of IDS.

The advantages of FPA are summarized as follows.

- FPA has a fast convergence rate that is exponentially better than other optimization algorithms [150].
- FPA can be applied for both single objective and multi-objective optimization problems [158].
- FPA algorithm has better exploration capabilities with local and global search properties [158].

The disadvantages of FPA are summarized as follows.

Table 10
Flower pollination algorithm for IDS.

Ref	Applicability	Parameters	Algorithm	Dataset	Dataset Size	Result	Highlights
[160]	Feature selection	Not specified	J48-NB-FPA	NSL-KDD	Training set: Not mentioned, Test set: 22,544	ROC analysis of the results obtained is shown.	• The pollination based feature selection method reduces the processing time. • Limited number of pollinators were used for feature selection process.
[162]	Feature set optimization	Not specified	NN-FPA	NSL-KDD	Not mentioned	Methodology to be implemented is proposed with flow chart and algorithm description.	• The hybridization of pollination algorithm improves the computational complexity of the classifier.
[161]	Feature selection	Not specified	SVM-FPA	KDD CUP 99	Not mentioned	ROC curve is presented showing a comparison of the approaches.	• The hybridization of FPA with SVM, results in improved accuracy as FPA classifies the data into linear and non-linear classes.

- The selection of switch probability is a challenging task as it controls the exploration and exploitation capability of FPA [160].
- The performance of the algorithm depends on the selection of appropriate global and local search techniques [161].

4. Applicability of Swarm and Evolutionary Algorithms in Intrusion Detection System

IDS is a challenging research area due to the enormous research being carried out in this field. Moreover, there are many IDS tools such as Snort [163], Suricata [164], Bro IDS [165], that are used to detect and prevent intrusions. However, the performance of these tools depend on the rules defined for detecting intrusions, also these tools are developed for specific platforms and for detecting a specific range of attacks. The tools require high computational resources for detecting attacks [166]. Therefore, there is a need to develop effective and efficient IDS models with high detection accuracy and reduced computational cost. Thus, the prospective research areas for developing IDS using SWEVO algorithms can be as follows: continuously evolving intrusions and vulnerabilities, collecting data for analysis, applying appropriate detection and response mechanism for addressing the attacks, studying about the infrastructure of the IDS, maintaining the security of IDS and evaluating the performance of IDS. The applicability areas of IDS where SWEVO algorithms are applied is shown in Fig. 12.

4.1. Classification of intrusions and vulnerabilities

Security can be stated as the internal asset of the software and this principle is followed by every organization [10]. Loopholes in the system lead to vulnerabilities that are exploited by the intruders to compromise the system. On the other hand unauthorized access, breach of security rule of an organization, injecting malicious code, and executing attack are the forms of intrusions that harm the system and network resources. To achieve this goal of compromising and harming the system, the intruders are constantly working towards creating new attack strategies by finding potential loopholes. To address the dynamic nature of continuously evolving intrusion scenarios, researchers are diligently working towards building new variants of existing algorithms using SWEVO algorithms. An evolutionary approach using GA and evolution strategies are developed to detect the buffer overflow vulnerability in C code [167]. In this work, the focus was to address the issue of dealing with constraints on string inputs that must be fulfilled to reach the vulnerable statement in the code. The approach learns constraints automatically by compiling the vulnerable code and dynamically generating the inputs for the same [167].

Mimicry attacks executed by exploiting the sequence of system calls automatically using the linear Genetic Programming (GP) [168]. In this method, the black-box technique is followed wherein the intruder has knowledge about the outcome of the IDS. Attack detection rate, attack success ratio, and attack length are used as the parameters of the fitness function. The attacks evolved with GP showed lower anomaly rates compared to original attacks [168]. The extended study of creating the mimicry attacks was implemented using the white box testing in which the delay parameter was added in the fitness function [169]. The results showed that the black box approach generates attacks with low anomaly rates than the white box. The evolutionary algorithms can also be used to address network vulnerability assessments which are the repository of the data analyzed by the security experts. An evolutionary multi-objective can be used to group the data of security assessments based on tested devices having similar vulnerabilities to identify the hidden patterns [170]. The SWEVO algorithms can also be used to derive variants of known attacks which make them a prospective method for dealing with unknown attacks [171]. In wireless networks, SWEVO algorithms can be used to detect and prevent the intruder from entering the network. For instance, ACO is used to detect stealthy

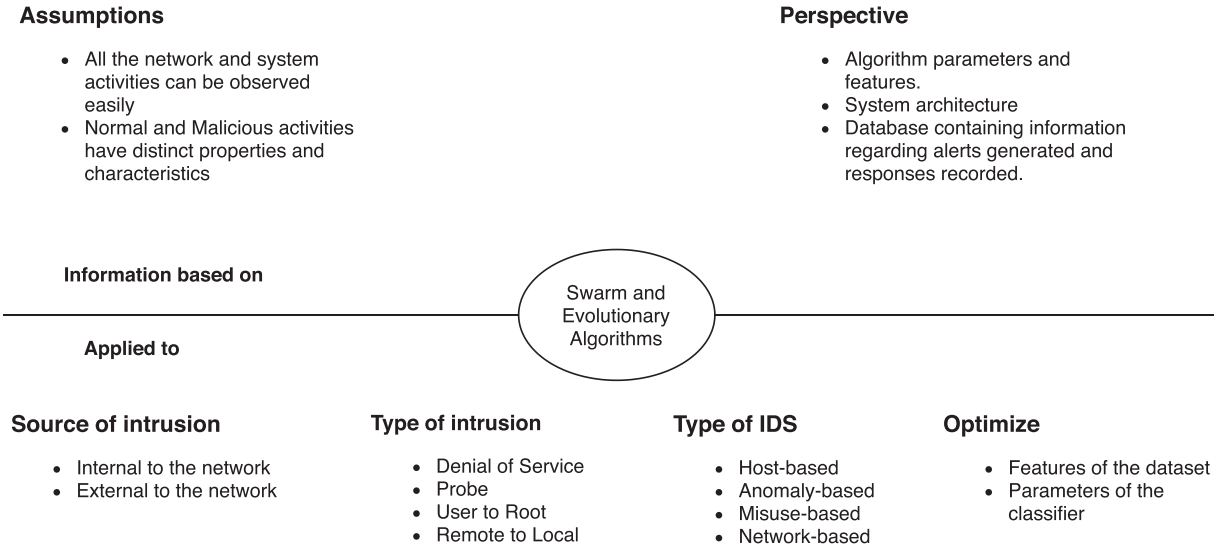


Fig. 12. Applicability of swarm and evolutionary algorithms in IDS

attacks and safeguard the nodes in the mobile ad-hoc network by preventing the attacker to enter the network [172].

Features of evolutionary algorithms like the representation of a possible set of solutions with ease and optimizing the problem using multi-objective methods make their use more appealing for evolving known and unknown attacks. One of the proposed methods is based on GP where the performance of the IDS is compared with the DT classifier [173] and support vector machine [95] using KDD CUP 99 dataset. The results showed that GP performs better than both techniques in classifying attacks. Linear GP is compared with SVM and neural network for detecting all kinds of attacks present in the KDD CUP 99 dataset. The linear GP showed significant results compared to both the methods but it requires a very high training time. The SWEVO algorithms are also applied for increasing the performance efficiency of the classifiers. For instance, PSO is used to optimize the performance of the neural network in Ref. [174] and to enhance the clustering activities in Ref. [175].

Similarly, as presented in Ref. [176], SVM performs equally well with GP, demonstrating the effects of the control parameters such as program size and variation operators on the results. A study of six classifiers namely multi-layer perceptron, a linear classifier, the Gaussian mixture model, the NB classifier, SVM, and GP, for detecting attacks like dropping, blackhole, flooding, and forging in ad hoc networks is presented in Ref. [176]. The simulation is carried out using the AODV protocol wherein different mobility levels were not considered. The results showed that SVM and GP outperform the other methods. It is mentioned that the GP is good for binary classification and SVM is good for multi-class classification of attacks.

4.2. Feature selection and optimization

SWEVO algorithms are used for selecting significant features from the dataset for enhancing attack classification process. While deploying IDS, selecting features ease processing large datasets. Feature selection process selects the prominent features that are useful in processing the dataset in a better way. Feature relevancy enhances the performance of the algorithms used for intrusion detection and also reduces the training time for the underlying classifier [177]. Features selection techniques select the features with minimum redundancy and maximum relevance score for detecting intrusions. The most conventional methods used for feature selection are filter and wrapper that eliminates irrelevant and redundant features to increase the accuracy of the model. SWEVO algorithms such as FA, ABC, BA, GA, ACO, and PSO are also used for feature selection.

The reduction in the dimension of the dataset holds significance in detecting anomalies from the high dimensional network traffic. For instance, FA is used as a feature selection method with DT classifier and Bayesian networks for selecting features from the KDD CUP 99 dataset [149]. The method states that only ten most significant features are required for improving the accuracy of the model [149]. The hybrid model of SVM with PSO delivers higher DR by choosing the optimum feature subset using the binary PSO method [112]. The features from the dataset can also be optimized using PSO to find the best feature subset and it can also outperform GA for searching the most significant features [116].

Binary Bat algorithm and binary PSO are used with SVM and DT classifiers to reduce the dimension of the features of NSL-KDD dataset [156] whereas ABC algorithm is used with Adaboost algorithm as feature selection method in Ref. [178]. The work proposed in Ref. [156] reduces the size of the KDD CUP 99 dataset to 64% and results showed better accuracy even for the unknown attacks. ACO tested on the KDD CUP 99 and NSL KDD dataset for eliminating superfluous features and deriving an optimum set of features for evaluation [98]. Furthermore, evolutionary algorithms can be used to eliminate costly features from the dataset that demand a large amount of time for computation. This can be advantageous when a large amount of network traffic is being monitored [98].

4.3. Intrusion detection and response mechanism

The most important part of IDS is choosing an appropriate detection technique for identifying attacks. Many techniques such as SVM, neural network, k-nearest neighbour, tree classifiers, and statistical techniques have been used to address the problem of intrusion detection. Intrusion detection requires easily adaptable methods with high fault tolerance and computation speed. Therefore, evolutionary algorithms are one of the promising approaches that can be applied for intrusion detection because of its lightweight decision rules and simple output which is easy to understand [179].

The techniques used for intrusion detection are the machine learning classifiers which can be used in combination to classify the attack. Such a combination of classifiers is referred to as ensemble classifiers. PSO generated weights are used to create an ensemble-based classification model along with weighted majority algorithm [180]. In this ensemble approach, KNN and SVM are combined to detect the attacks from the KDD CUP 99 dataset. PSO is a global optimization algorithm that can be used to improve the performance of any classifier [12]. Evolution-

ary algorithms can be used to optimize the weights of the model. For example, the artificial bee colony algorithm was used to optimize the values of the linkage weights and bias of the multilayer perception in Ref. [130]. In Ref. [181], FA is used to optimize the parameters of the outlier detection algorithm for high dimensional datasets.

The evolutionary algorithms can also be used for deriving better classification rules of the classifiers, for instance, ACO is used to generate high-quality fuzzy classification rule in Ref. [86]. ACO is inspired by the behaviour of ants in finding the shortest path to the food source. This analogy can be used to communicate in the distributed IDS for monitoring and exchanging detailed information [182]. Multi-objective evolutionary algorithms are used to give a set of outcomes that can provide trade-offs between the false negatives and false positives. GNP can be used for dealing with misuse and anomaly-based detection [60].

4.4. IDS infrastructure

The placement of IDS in the network environment also holds importance for detecting the attacks. Some studies on evolutionary algorithms inspect the placement and infrastructure of IDS in the network. Also, multi-objective evolutionary algorithms take into account various objectives while establishing the IDS. GAs are used for the placement of IDS sensors by conforming to the conflict objectives [183]. The high false alarm rates have a negative impact on the detection rates of attacks. Here, the assumption is made that the placement of IDS for different attacks is different. For instance, the placement of IDS sensor for probing attack scenarios is not suitable for denial of service attack scenarios [183]. The paper focussed on four things namely, the number of sensors, false alarm rate, detection rate, and analyzing the costs calculated based on network traffic being monitored [183]. All the nodes in the network exhibit the same properties like detection capabilities and handling risk during the execution of the experiment but the placement of IDS sensors have different trade-offs.

There are different infrastructures of IDS such as stand-alone, distributed, and hierarchical. The multi-objective evolutionary algorithms work efficiently for an ad-hoc network having a distributed infrastructure of a node. As the stand-alone infrastructure contains nodes with local monitoring and detection ability, the hierarchical and distributed infrastructure should be taken into consideration for detecting attacks on a larger scale. In hierarchical infrastructure, the nodes are placed in a tree-like structure and each node is capable of detecting and analyzing the attack. These nodes communicate with each other in a bottom-up manner to collect all the necessary information for attack detection and communicate the same to the other nodes. A distributed and collaborative infrastructure for defining IDS for resource-constrained environments is presented in Ref. [18]. Here, the results showed improved classification accuracy by utilizing fewer number of nodes as compared to stand-alone infrastructure. The classification accuracy presented in terms of detection rate and false alarm rate while the number of nodes denotes the use of bandwidth.

The application of evolutionary algorithms for discovering the complex properties of ad-hoc networks is presented in Ref. [184]. Here, GAs used for optimal node selection problem. The nodes are structured in a form of cluster-tree based on the energy consumed, quality of each node, area covered, and delay. Each node designed to perform distinct roles to optimize the objectives. An extended work for this is presented in Ref. [185] where GA is implemented in a two-phase as hybrid algorithm. In the first phase nodes with high responsibilities are defined while distinct roles of the nodes are defined in the second phase. The statistical results imply that the hybrid algorithm outperforms the GA in execution time for detecting intrusion in resource-constrained wireless networks [185]. GAs can also be used for misuse based IDS where the security of the monitoring node and node coverage are taken into consideration [63]. GA uses the weighted fitness function for evaluating the individuals which effectively detects the affected and compromised nodes.

4.5. Security of IDS

Apart from securing the network from the different attacks, IDS itself can be prone to attacks which can be listed as evasion, over-stimulation, poisoning, denial of service, reverse hijacking, and reverse engineering [186]. Evolutionary algorithms focus on evasion attacks on IDS [169], Also, GP can be used to implement reverse engineering for IDS behaviour [187]. The method proposed in Ref. [187] is based on the assumption that the intruder is unaware of the internal behaviour of the IDS. Here, the depth of the GP trees is kept to five. Easy to understand IDS models were built to generate evasion attacks against the DT based IDS. Machine learning techniques have been proposed for IDS and therefore, the focus here is to address adversarial machine learning techniques. The study of adversarial capabilities of IDS is carried out in Ref. [188]. Basically, IDS consists of nodes which can monitor the network traffic and detect any unusual behaviour. These nodes are distributed for aggregating and correlating the data for detecting distributed attacks. The proposed work does the empirical analysis of IDS to measure the risk and provides countermeasures based on the analysis performed. Minimizing the negative influence of the developed security mechanism on the quality of service of an industrial control system is considered in Ref. [189]. The paper aims to measure the strength of IDS and quality of service needed for the real-time operating system. In this study, BA is used for generating a set of optimal solutions that ensures the security of the IDS.

4.6. Digital ants: A case study

Digital Ants is a collaborative research program of Pacific Northwest national laboratory (PNNL) with Wake Forest University [190]. The research program aims to establish a focused strategy for adaptive defense and predictive modeling of cybersecurity issues. The motivation of developing Digital Ants software was incorporating real ant behvaiour for defending collaborative networks against malware [190]. Digital ants are formed through tiny programs based on the ant behaviour that wander throughout the system investigating for malicious programs such as viruses and worms or any unusual activity. This ant-based system enables many organizations to build a collective cyber defense mechanism that proactively responds to the attacks and retains the security and privacy of the system [191].

The ant-based framework consists of 3000 different types of digital ants such as sentinel and sergeant. The sentinel ant is the autonomic manager of digital ants that are gathered on an individual computer [190]. This ant performs various tasks such as to collect information from the ants, to control the state of the localhost, and to take necessary action if needed. The sergeant is also an autonomic manager of multiple sentinels. The number of sergeants used determines the dimension of the network. These ants also interact with human technicians [191].

The ants wander around the network, investigating activities such as network statistics or routing table information. If any unusual activity is detected, it leaves a pheromone trail that attracts more ants towards the same machine on which the intrusion is detected. Different ants provide different information and based on these chunks of information a ground-base for the intrusion can be deduced. Unlike, an anti-virus program that runs scans for finding malicious software, this approach is based on the ant population that deviates based on threat level [190]. In a digital ant framework, initially, a base population is defined. If an ant is successful in finding an intrusion than it is replicated, if not that it dies. Thus, in this way the base population is maintained.

5. Challenges and Future Scope

Over the past decade, SWEVO algorithms based IDS have been extensively studied for building reliable and intelligent IDS models. The SWEVO methods have been combined with various machine learning techniques for designing IDS. For instance, SWEVO algorithms com-

bined with machine learning based techniques such as neural networks, SVM, NB, DT, random forest, k-means clustering, and fuzzy logic to enhance the performance of the classifier by optimizing the parameters used by the classifier. SWEVO algorithms are also used for feature optimization. These methods are also capable of recognizing known and unknown patterns to easily comprehend the intrusive behaviour and predict the uncertainty and imprecision of network activities, and thus, improving the ability of an IDS. As per the study presented in this paper, these techniques have been widely used for optimizing the parameters of the classifier for attack classification, optimizing feature set, and feature selection.

The applicability of SWEVO algorithms has shown that these methods are either hybridized or ensembled to develop an efficient IDS with significant detection rate and accuracy in real network scenarios. These methods segregate a large problem into smaller problems by assigning individuals to work in coordination, adaptive, and cost-effective environment. Even though favorable outcomes have been recorded by the SWEVO algorithms for enhancing the performance of IDS, there exist certain challenges that may be considered for future research in this field.

- The combination of machine learning techniques with SWEVO algorithms is being profusely used to expedite the performance of the classifier. However, choosing appropriate parameters is a challenging task for performing either parameter optimization or feature selection.
- For performance evaluation of the method being proposed, intrusion detection dataset plays a key role. There is a need for updating the datasets for IDS. The datasets such as KDD CUP 99 and DARPA which have been used so far lack the realistic network scenarios for IDS [192]. An IDS model built using these datasets may perform differently in real network scenarios. This dataset consists of some properties such as large dimension and volume with skewed data distribution. For instance, DARPA dataset exhibit irregularities in network traffic with large differences in TTL for attacks classes and normal traffic that gives high-performance rate even for simple IDS models. KDD CUP 99 dataset has an uneven distribution of data samples among the defined attack categories, and therefore, evaluating the performance of classifier based on this dataset alone is not sufficient [192]. Thus, there is a high demand for developing a dataset for IDS with real network scenarios by considering packet payload as well as packet header information.
- Performance of a classifier depends on the quantity and quality of features used for attack detection and classification. Therefore, establishing the relationship between the network flow features and attack types is a challenging task. The computational complexity increase with an increase in the number of network features. Whereas, if less number of features are selected, then there is a probability of missing out essential network information. Hence, it is necessary to explore the role of network features in the performance of IDS.
- Another challenging aspect of intrusion detection is adaptability to continuously changing network scenarios. The network environment consists of intrusion activities that keep on changing over time. In addition, network system and user actions that evolve over time. The IDS should have the capability to adapt to the behavioural changes to maintain the detection accuracy of the system.
- An IDS should be able to handle a large amount of data that makes it difficult to build an efficient IDS. For instance, the dataset KDD CUP 99 lists about 50,00,000 data samples with 41 features. Thus, any learning algorithm should have characteristics of self-adaptability and parallelism to handle a huge amount of data.

The techniques discussed in the survey have their genesis in biology and therefore, placing an analogy between the algorithms and biology is relatively simple. The algorithms studied for intrusion detection have applied a detailed understanding of biotic behaviour of individuals. The in-depth study and applications of these nature-inspired algorithms in the field of intrusion detection are the subject of future research work concerning the following areas.

- There is an increase in end-to-end encryption in network communication. Therefore, research addressing encrypted network traffic should be performed. The techniques developed should be able to extract network information from the encrypted packet for analysis.
- Intrusion detection dataset consists of flow-based network features and packet-based network features. Therefore, a review on the performance of the techniques based on flow-based network features and packet-based network features should be carried out.
- Moreover, the dataset used for the performance evaluation suffers from class imbalance problem. Hence, SWEVO-based techniques should be designed to address this problem.
- SWEVO techniques to address alert correlation is the subject of future research work for reducing the false alarm rate in IDS.

6. Conclusion

The role of the SWEVO algorithms in the field of intrusion detection has been the center of attraction for the research community over the decade. The characteristics of these methods such as self-adaptability, resistant towards faults, computational efficiency, and handling errors generated due to unwanted noise in the data help to build IDS models. The paper surveys state-of-the-art methods in the field of swarm intelligence and evolutionary computation from the year 2008–2018. The discussion on the scope of these methods and their applicability in the field of IDS is the objective of this survey. These algorithms are capable of generating easily decodable results for security experts [9]. Moreover, they generate lightweight outcomes and produce results that represent the relationship between the conflicting parameters. The use of these methods in improving the performance of ML classifiers is also discussed in the paper. These methods are capable of combining the advantages of the ML algorithms to deliver better solutions in terms of detection rate, accuracy, and false alarm rate. Section 1 gives a brief introduction to IDS and discusses related work performed in the field of IDS using SWEVO algorithms. Section 2 gives an overview of SWEVO algorithms, with a discussion on the dynamics of SWEVO algorithms and their mechanisms. In section 3, a literature review on various SWEVO-based IDS techniques are described with their working, control parameters, fitness function, and results. The applicability of SWEVO algorithms in the field of IDS is discussed in Section 4. The challenges and potential future research directions for IDS based on SWEVO algorithms are highlighted in section 5. It is essential to note that all the algorithms strive to attain a good optimal solution that can be achieved by parameter optimization, feature optimization, and attack classification. The paper summarizes the methods systematically along with highlighting the advantages and disadvantages of the techniques. Thus, this survey can be considered as a manifesto for novice researchers working in the field of SWEVO algorithms and IDS.

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